

BFS Traversal

Step 1: Start at A

Legend

Complete

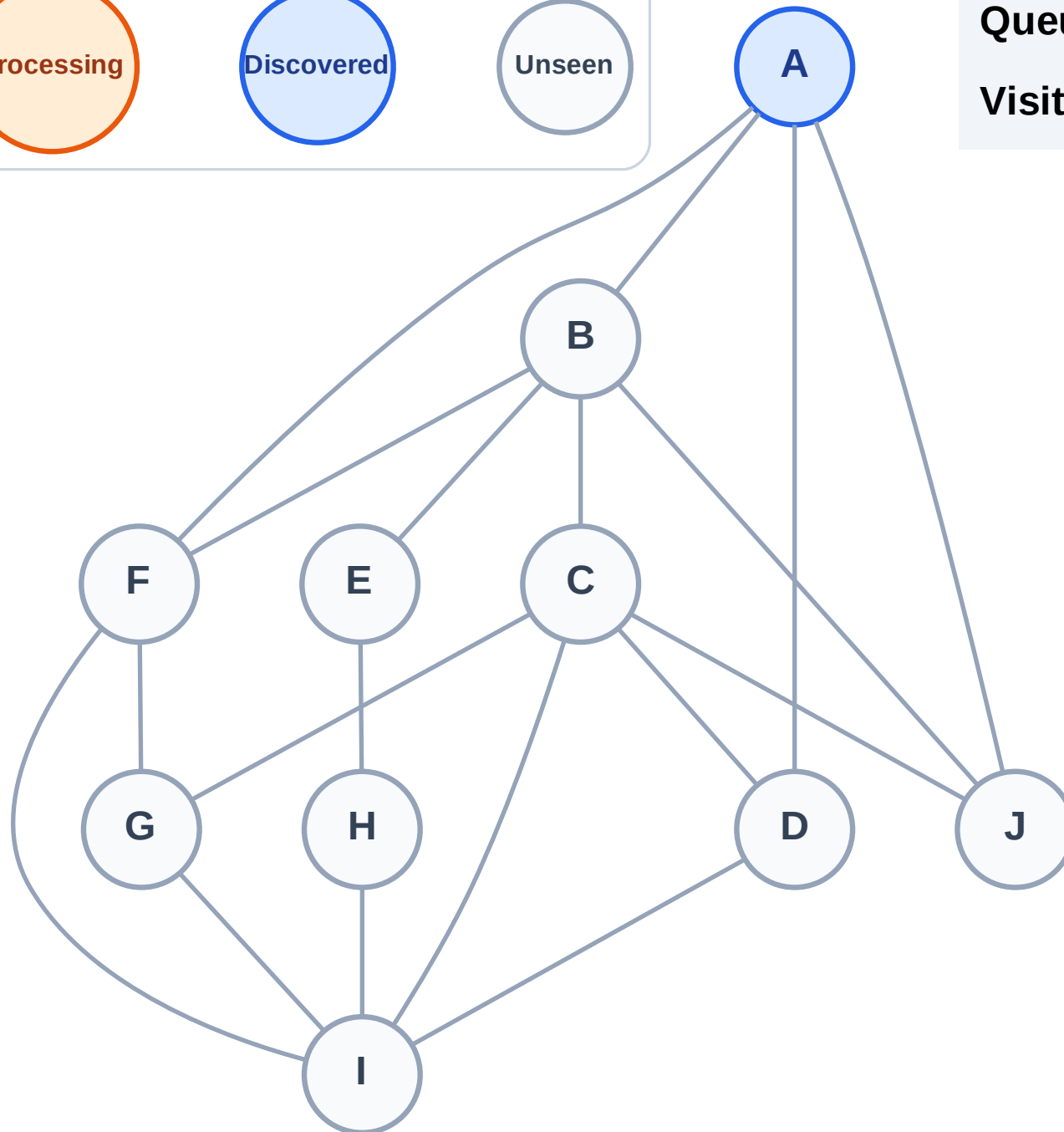
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

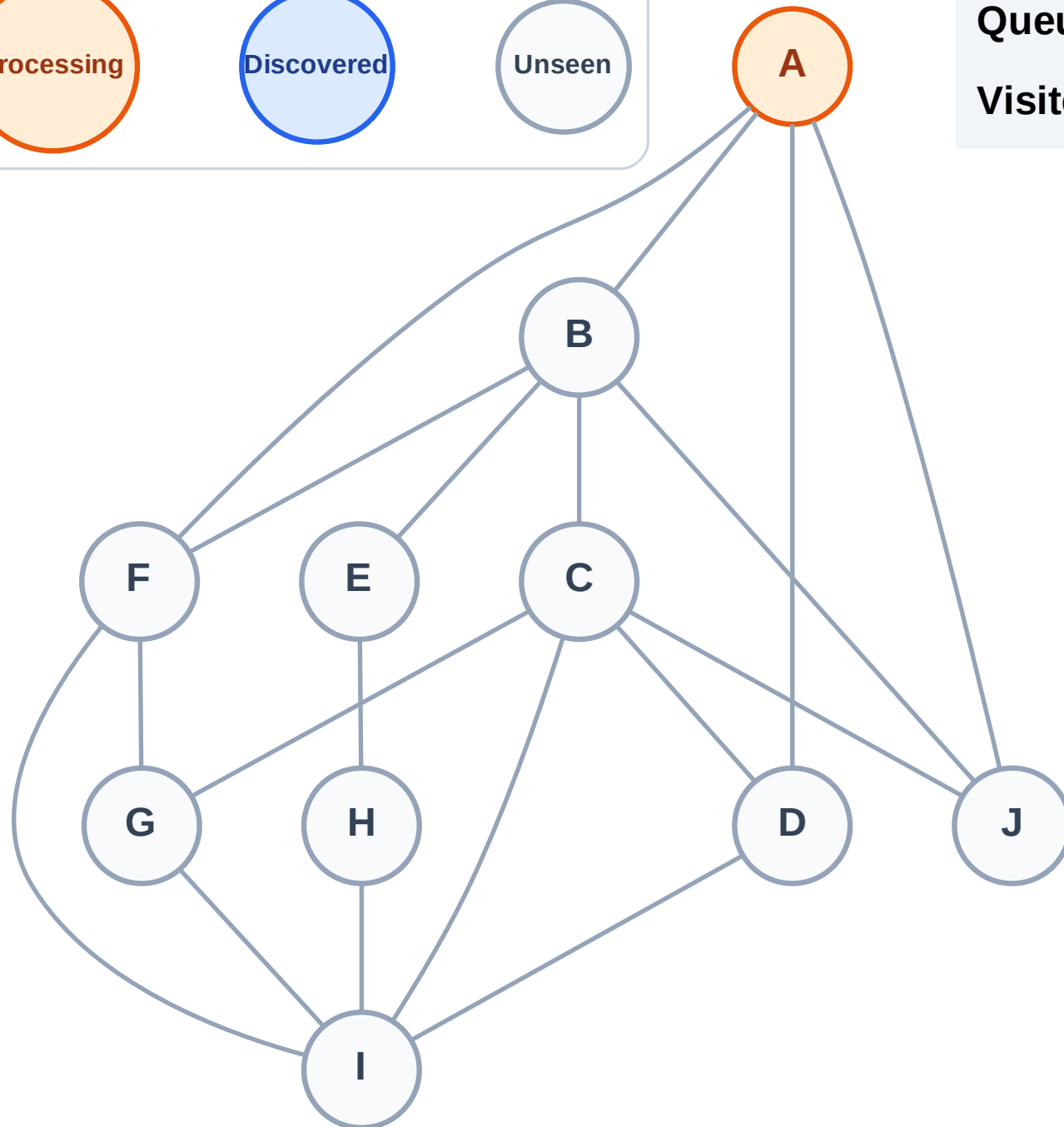
Step 2: Process node A

Legend



Queue: (empty)

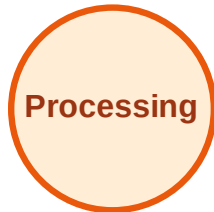
Visited: (none)



BFS Traversal

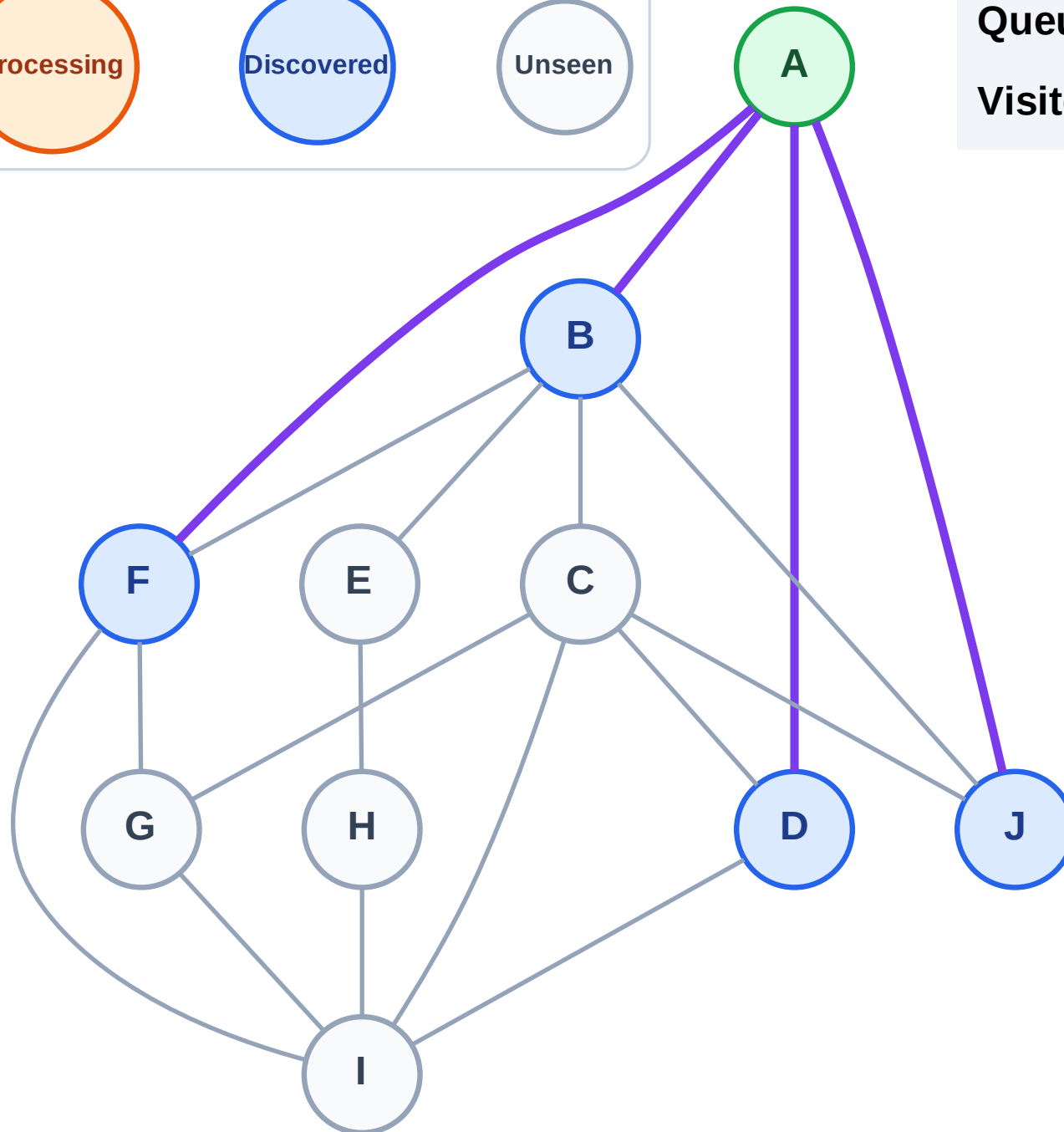
Step 3: Discover B, D, F, J from A

Legend



Queue: B → D → F → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

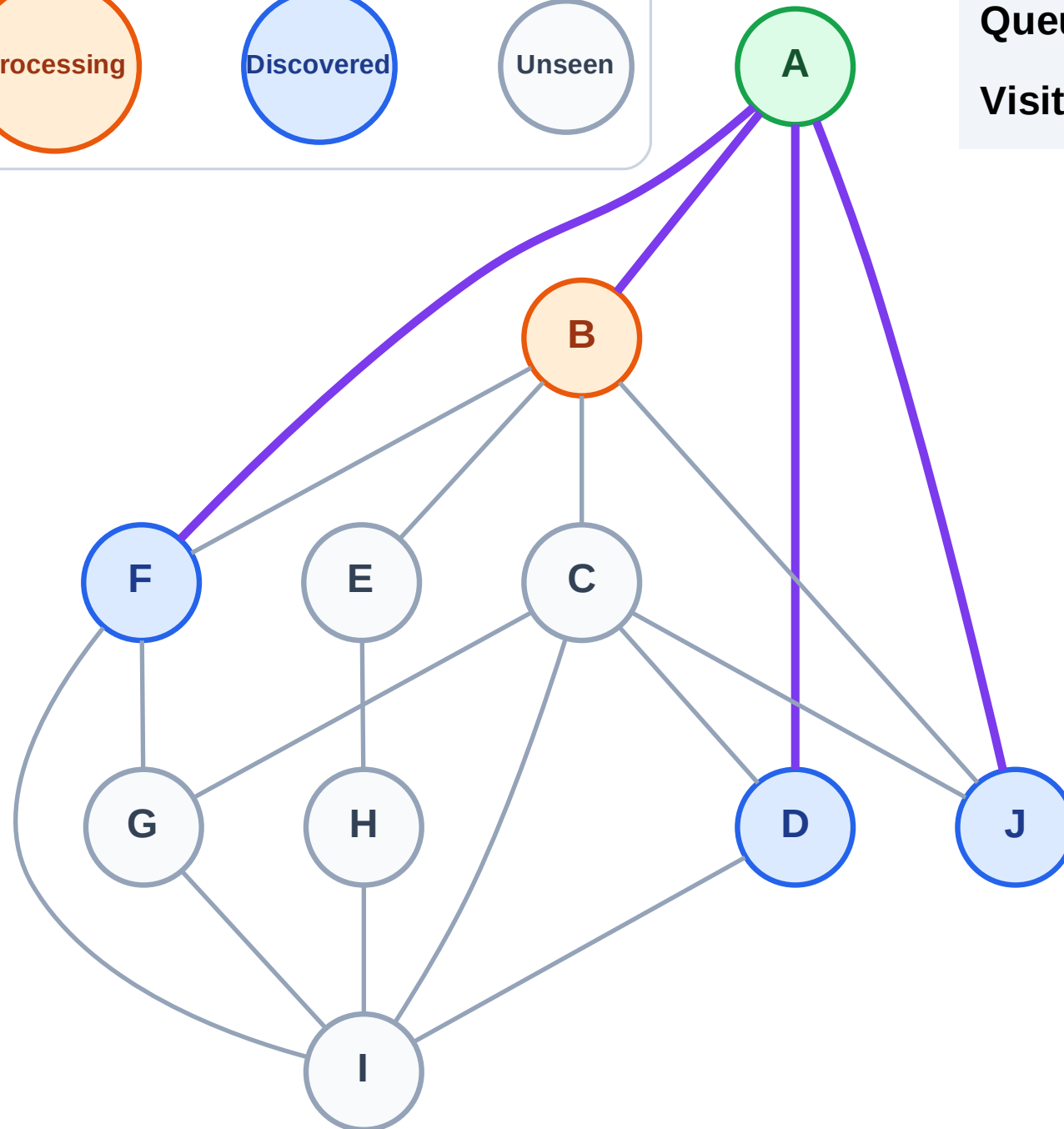
Processing

Discovered

Unseen

Queue: D → F → J

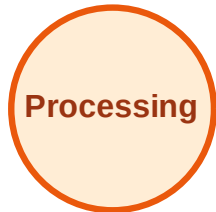
Visited: A



BFS Traversal

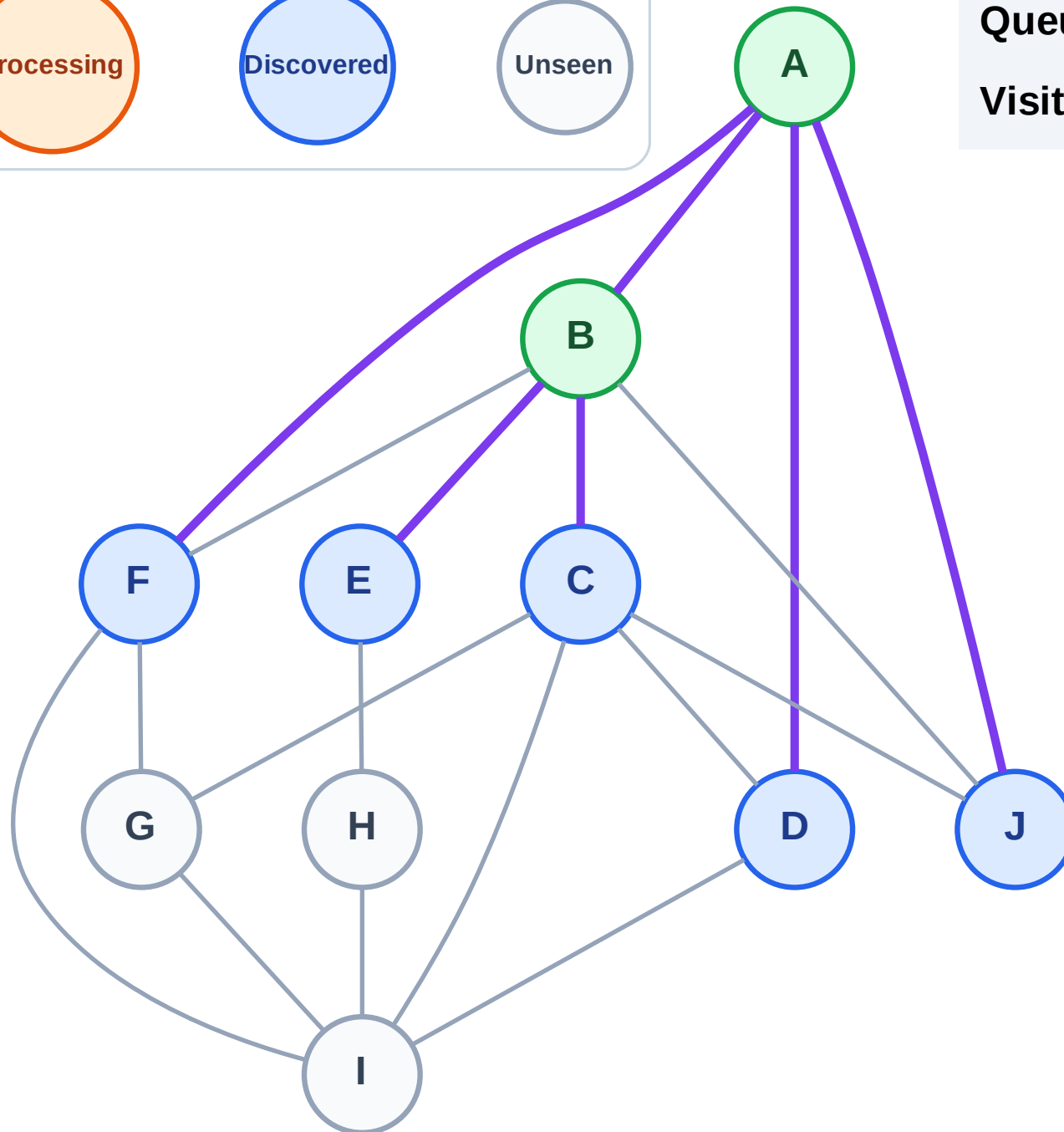
Step 5: Discover C, E from B

Legend



Queue: D → F → J → C → E

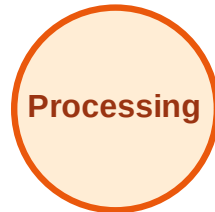
Visited: A, B



BFS Traversal

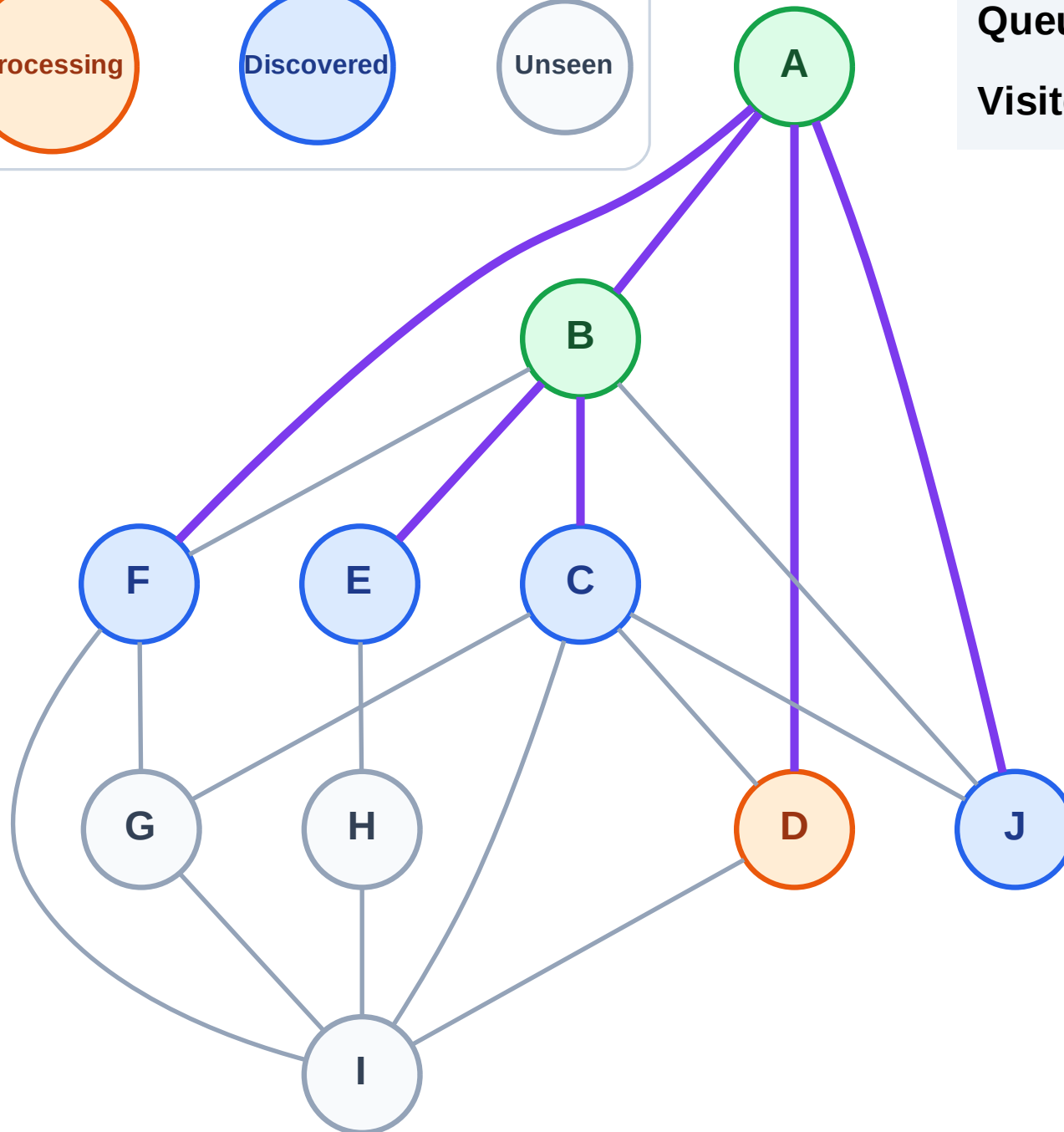
Step 6: Process node D

Legend



Queue: F → J → C → E

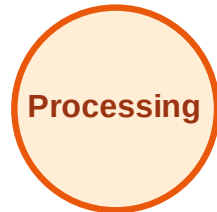
Visited: A, B



BFS Traversal

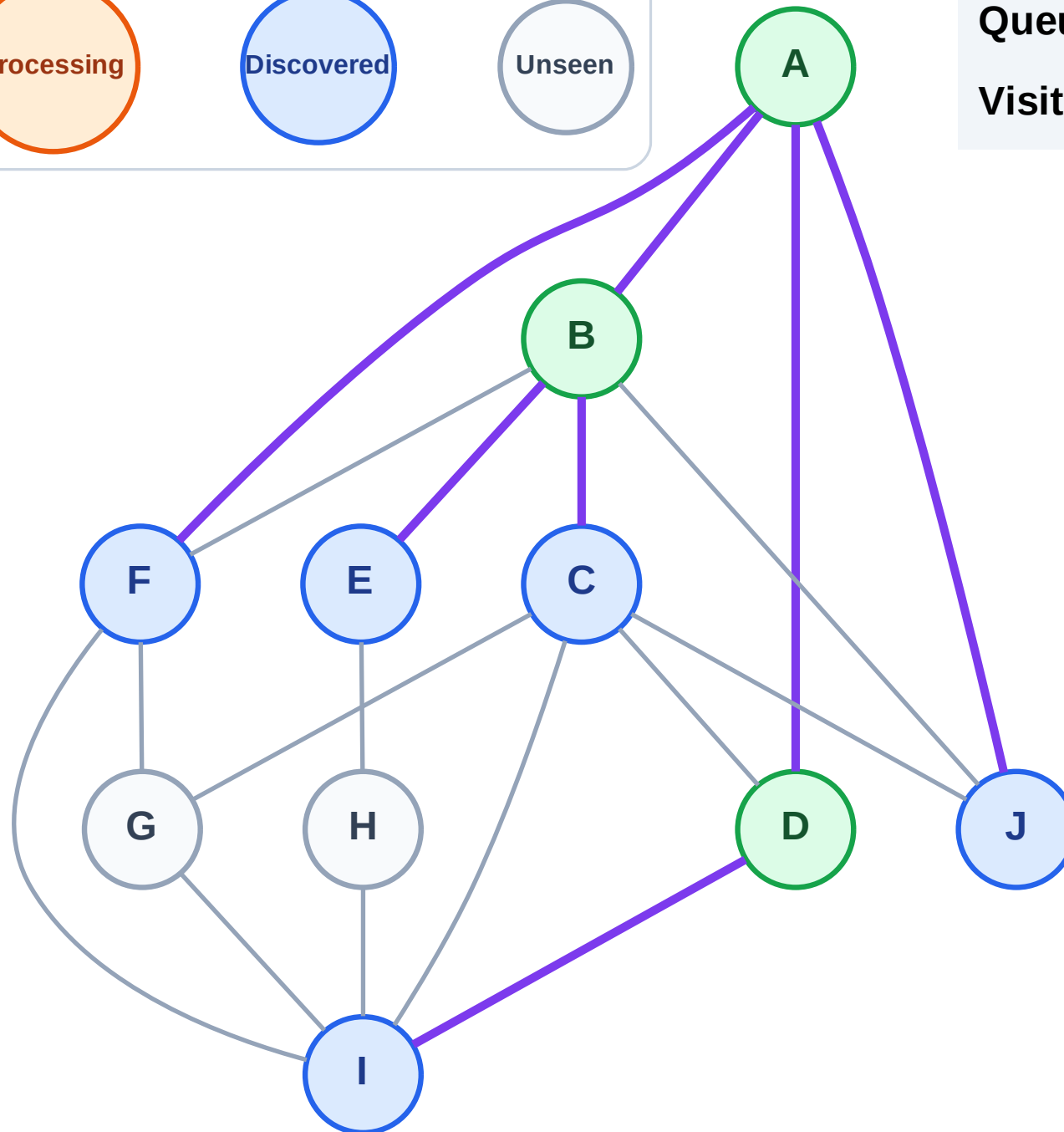
Step 7: Discover I from D

Legend



Queue: F → J → C → E → I

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

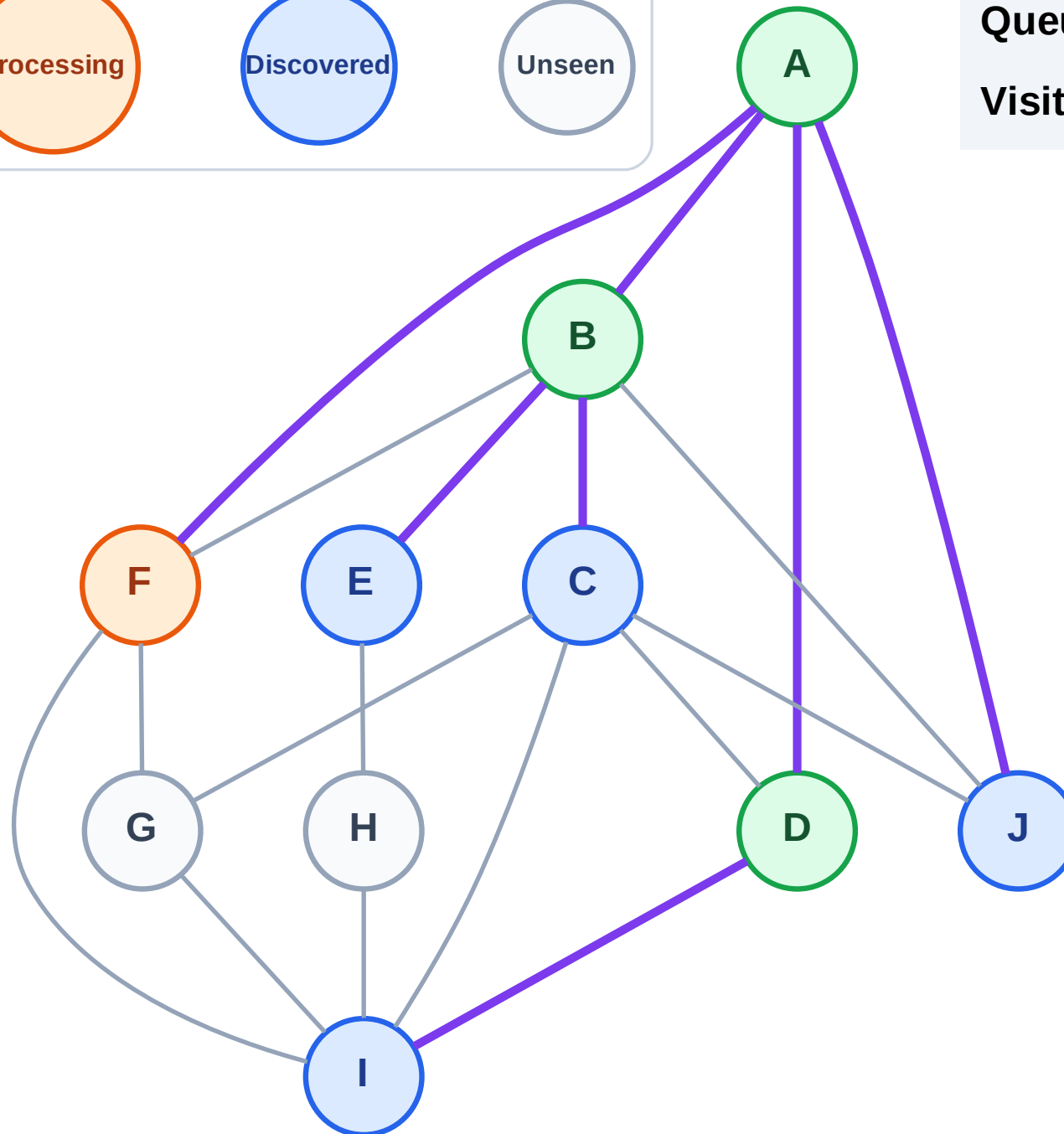
Processing

Discovered

Unseen

Queue: J → C → E → I

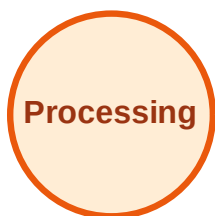
Visited: A, B, D



BFS Traversal

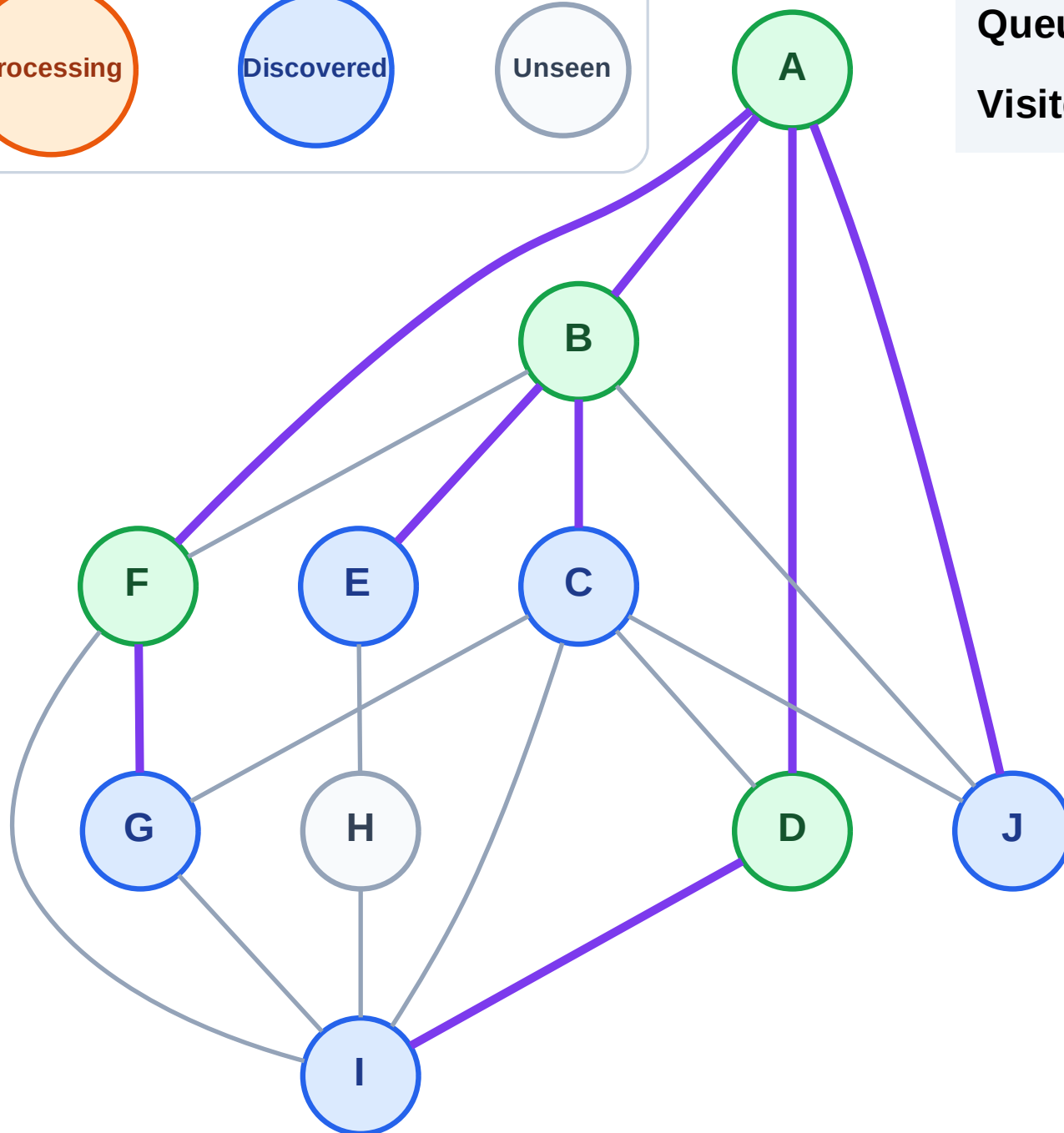
Step 9: Discover G from F

Legend



Queue: J → C → E → I → G

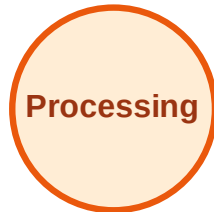
Visited: A, B, D, F



BFS Traversal

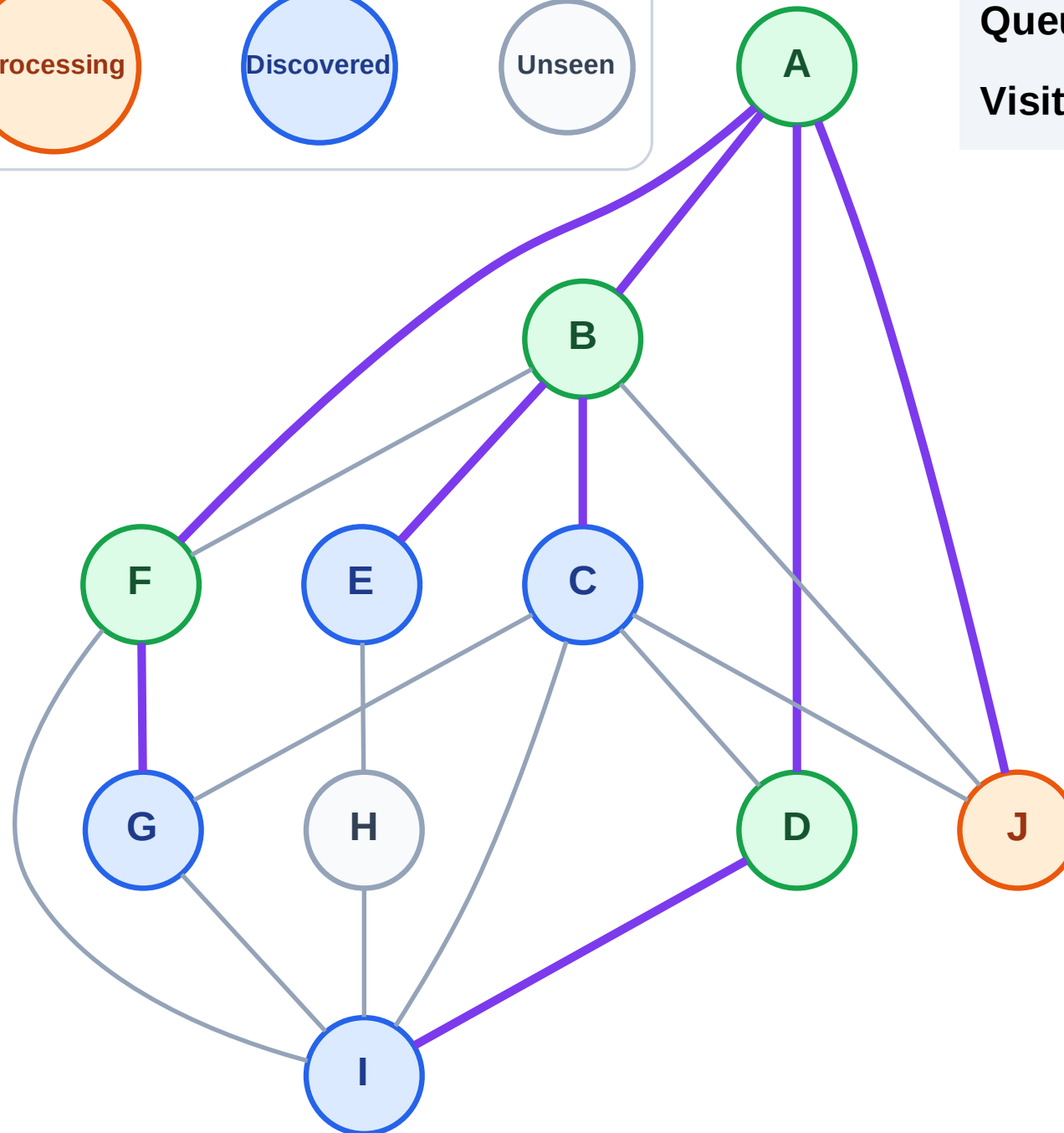
Step 10: Process node J

Legend



Queue: C → E → I → G

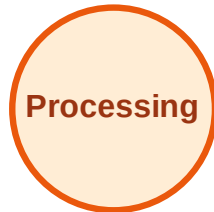
Visited: A, B, D, F



BFS Traversal

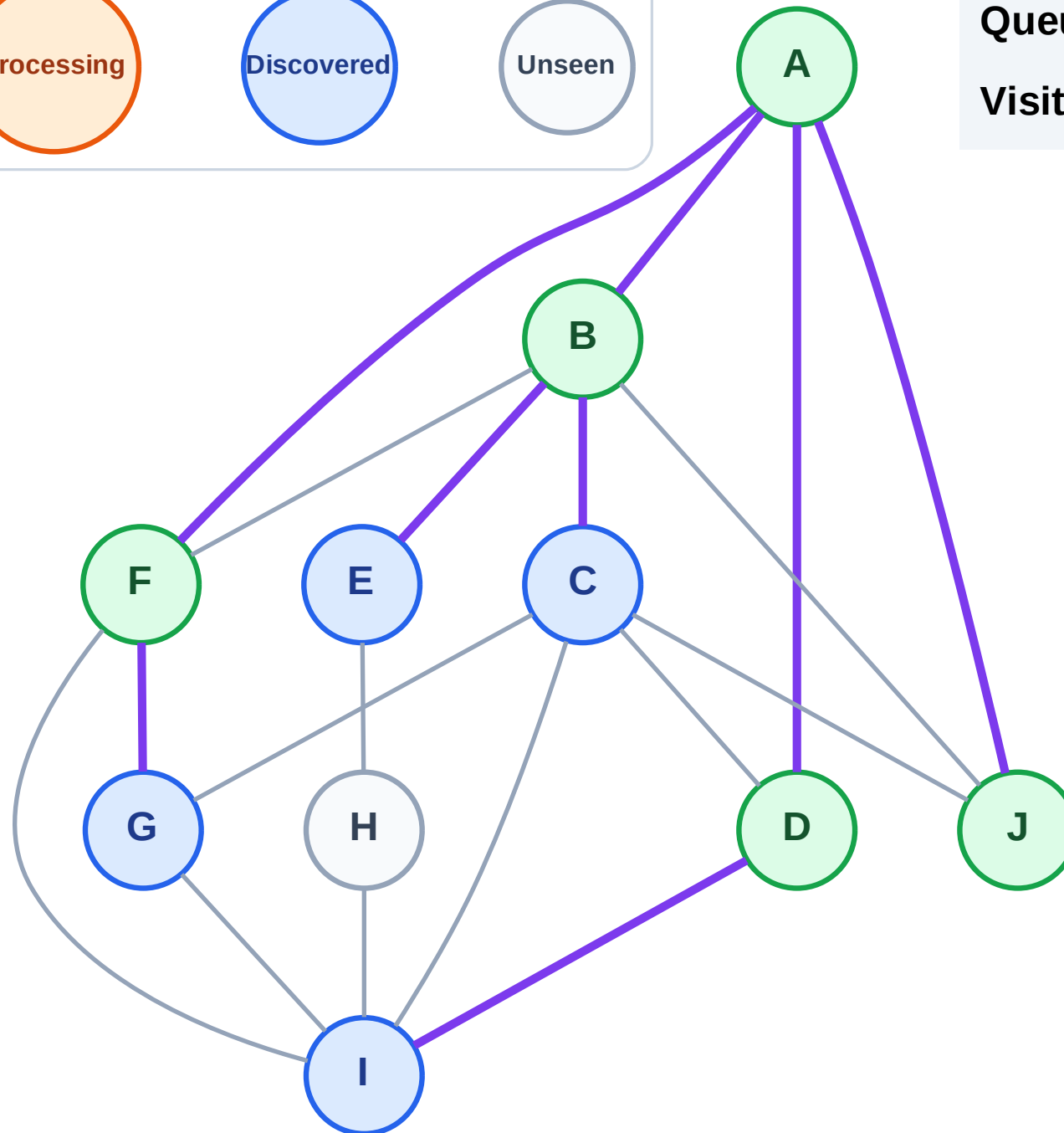
Step 11: No new neighbors from J

Legend



Queue: C → E → I → G

Visited: A, B, D, F, J



BFS Traversal

Step 12: Process node C

Legend

Complete

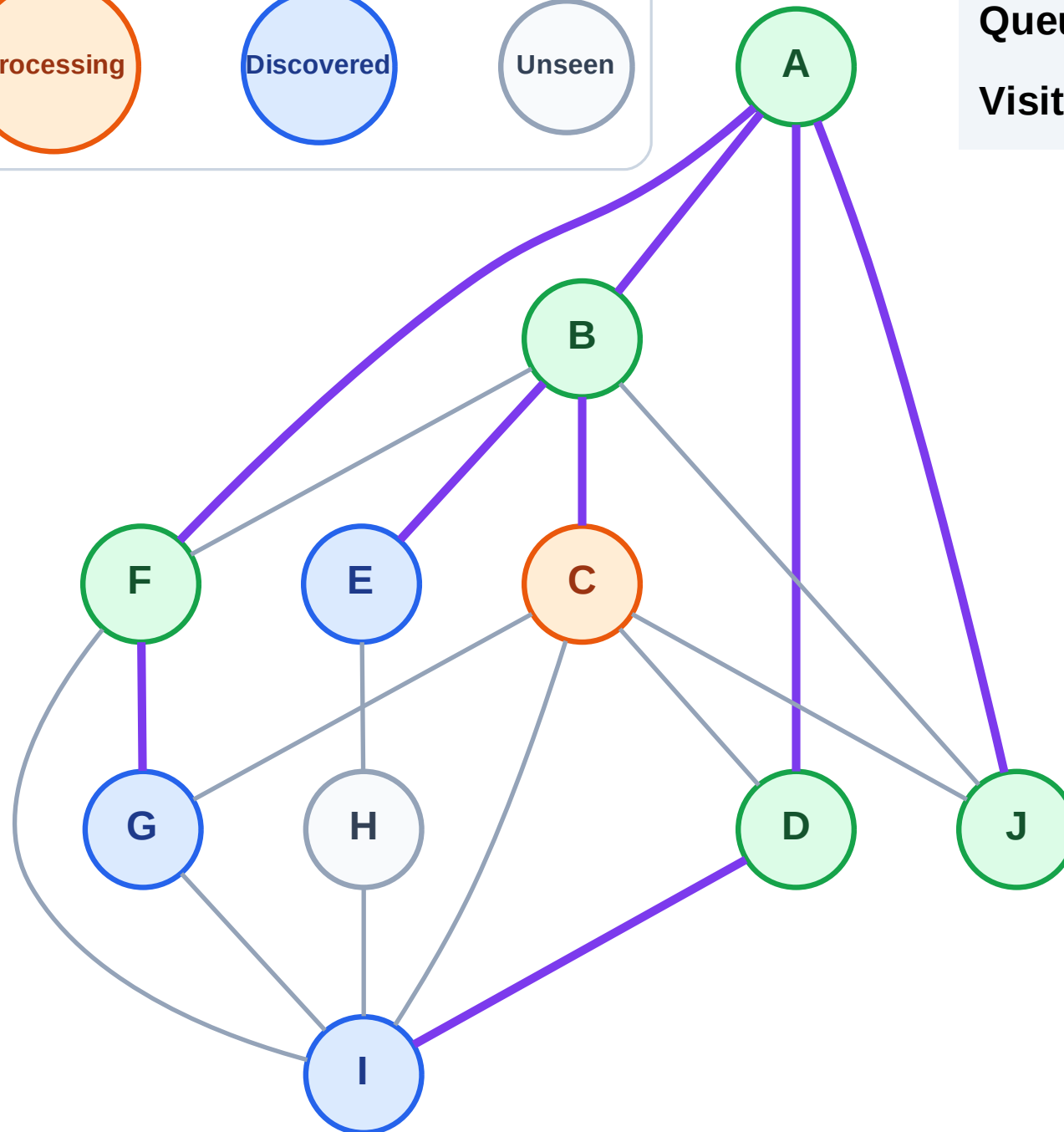
Processing

Discovered

Unseen

Queue: E → I → G

Visited: A, B, D, F, J



BFS Traversal

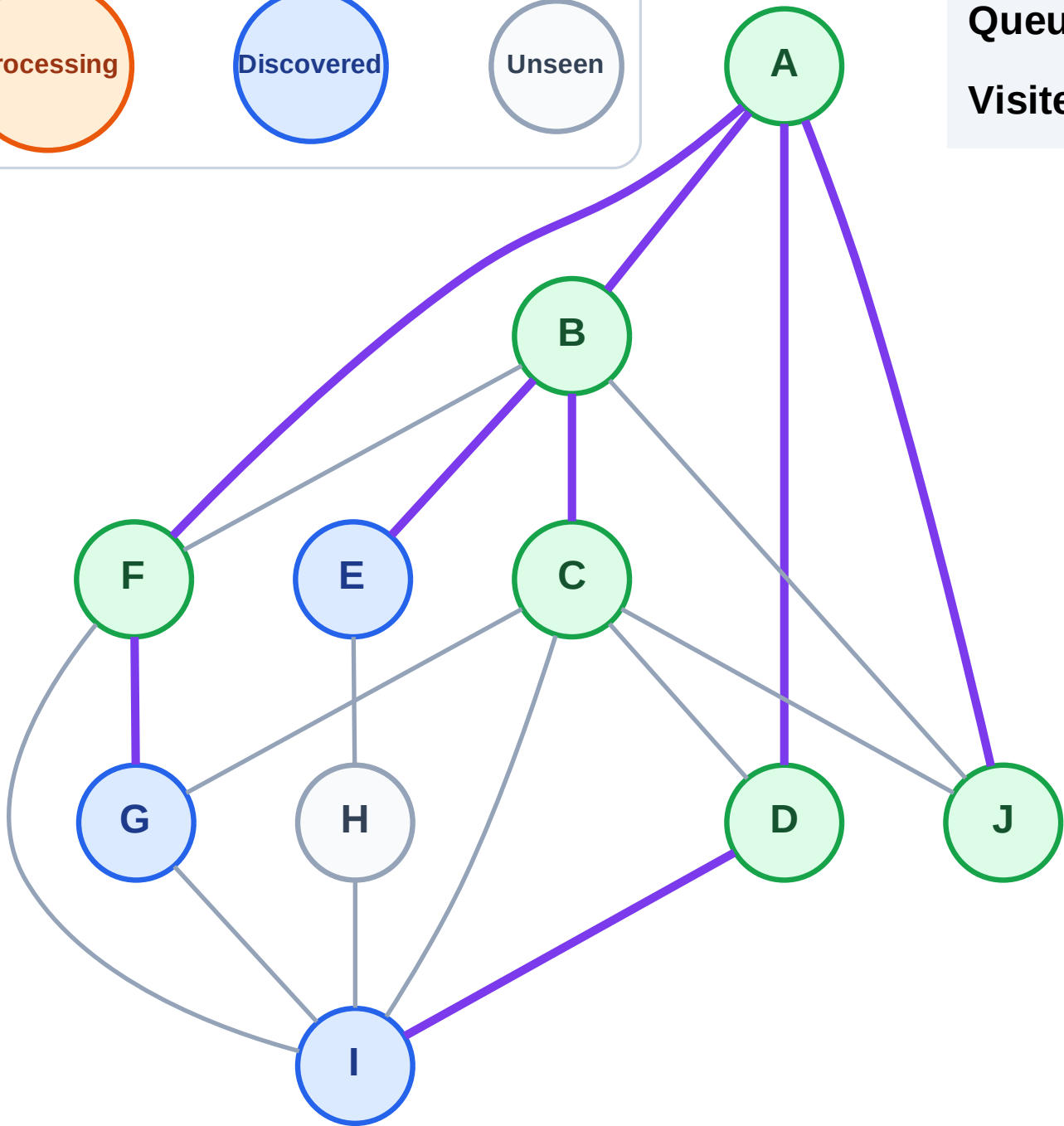
Step 13: No new neighbors from C

Legend



Queue: E → I → G

Visited: A, B, C, D, F, J



BFS Traversal

Step 14: Process node E

Legend

Complete

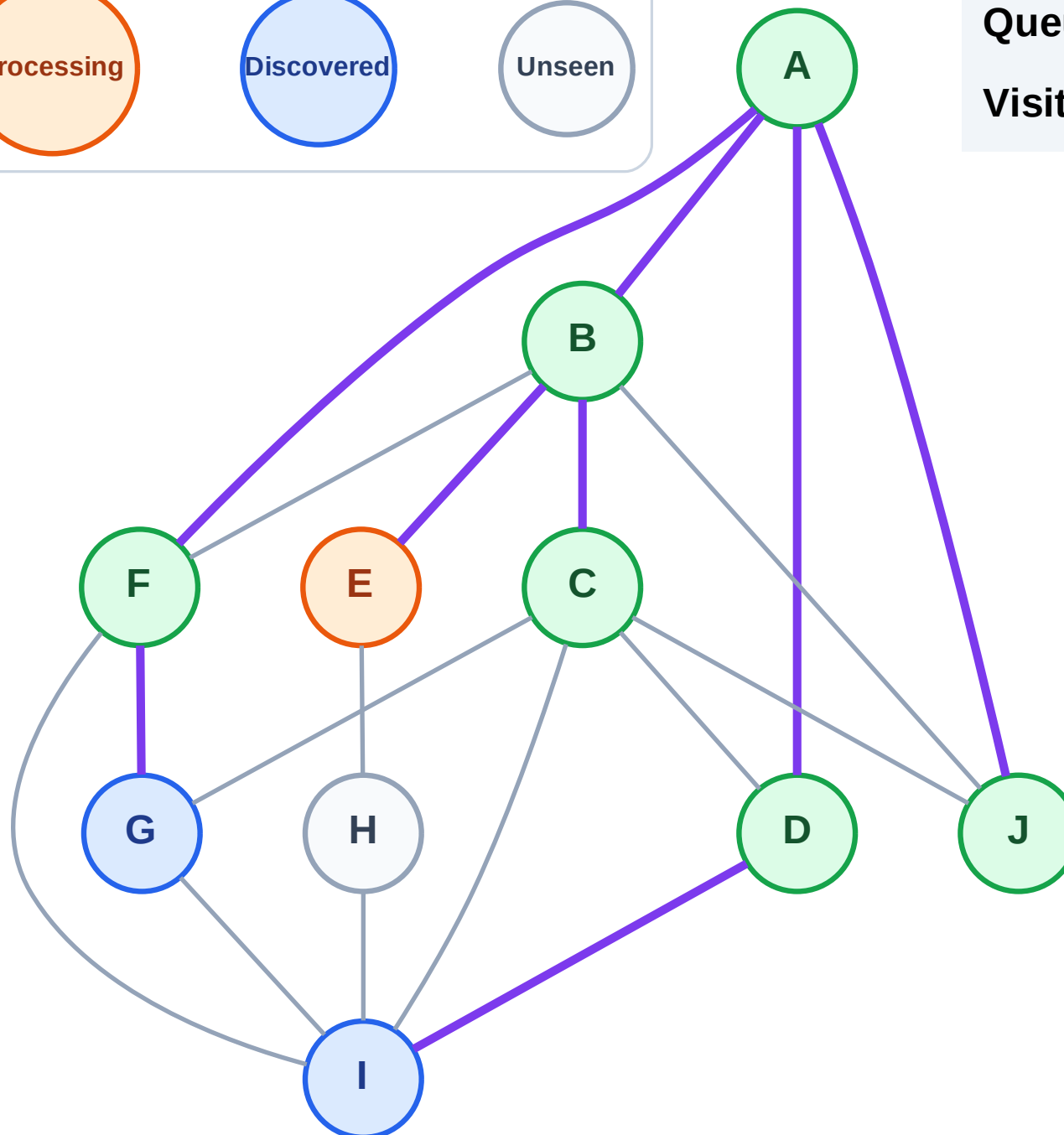
Processing

Discovered

Unseen

Queue: I → G

Visited: A, B, C, D, F, J



BFS Traversal

Step 15: Discover H from E

Legend

Complete

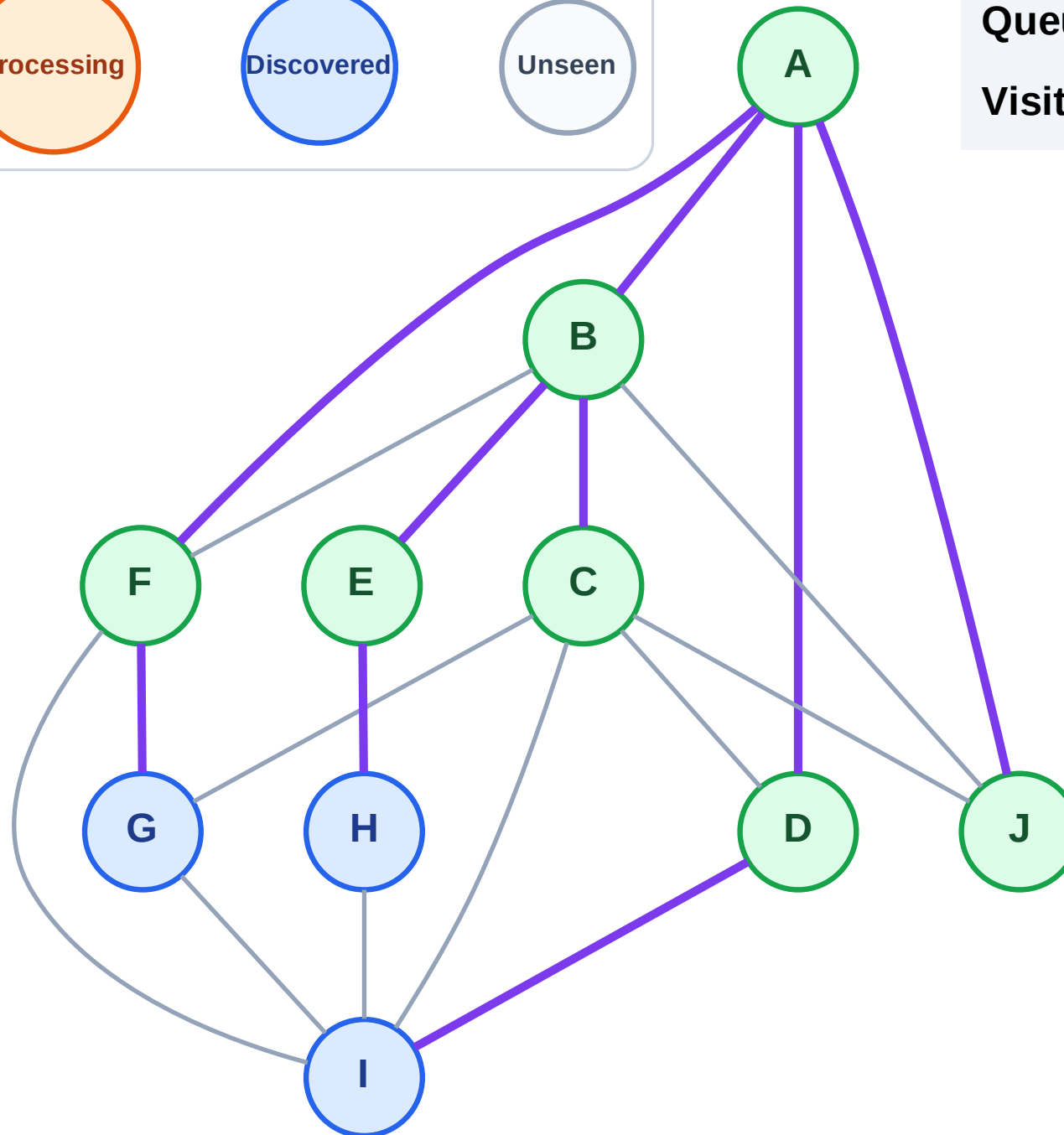
Processing

Discovered

Unseen

Queue: I → G → H

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 16: Process node I

Legend

Complete

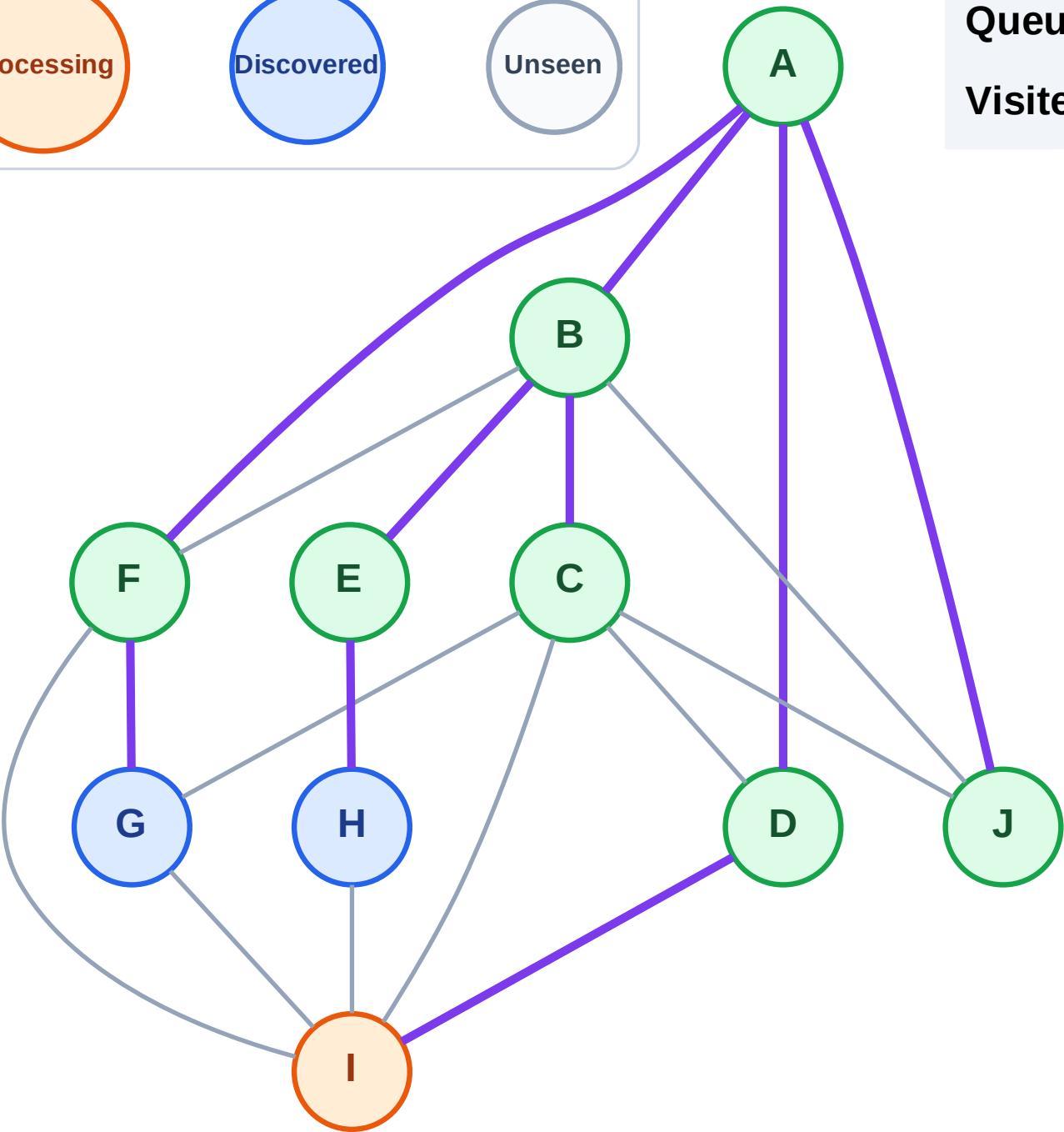
Processing

Discovered

Unseen

Queue: G → H

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 17: No new neighbors from I

Legend

Complete

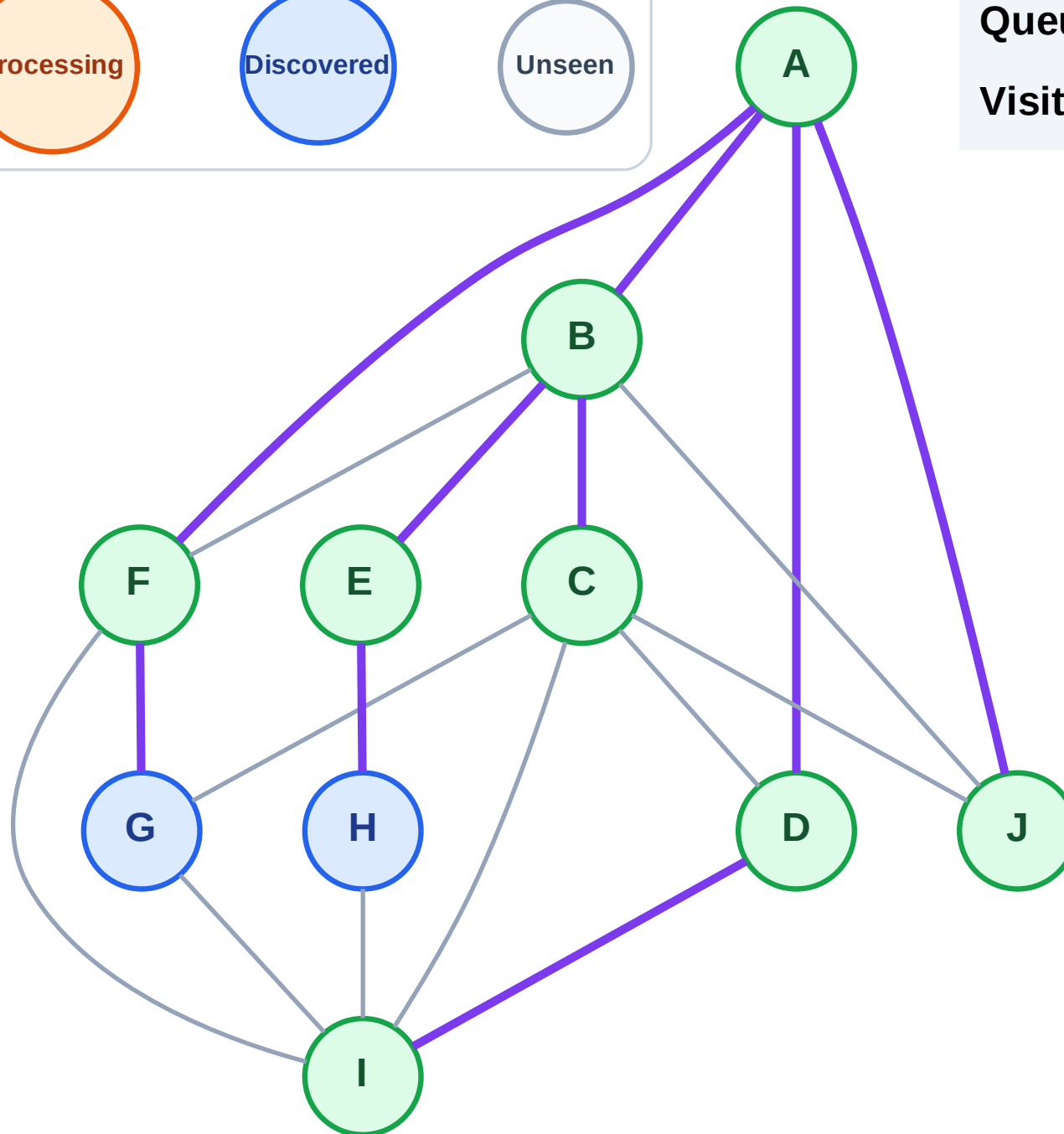
Processing

Discovered

Unseen

Queue: G → H

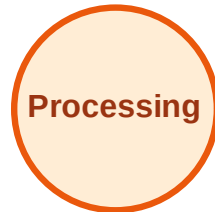
Visited: A, B, C, D, E, F, I, J



BFS Traversal

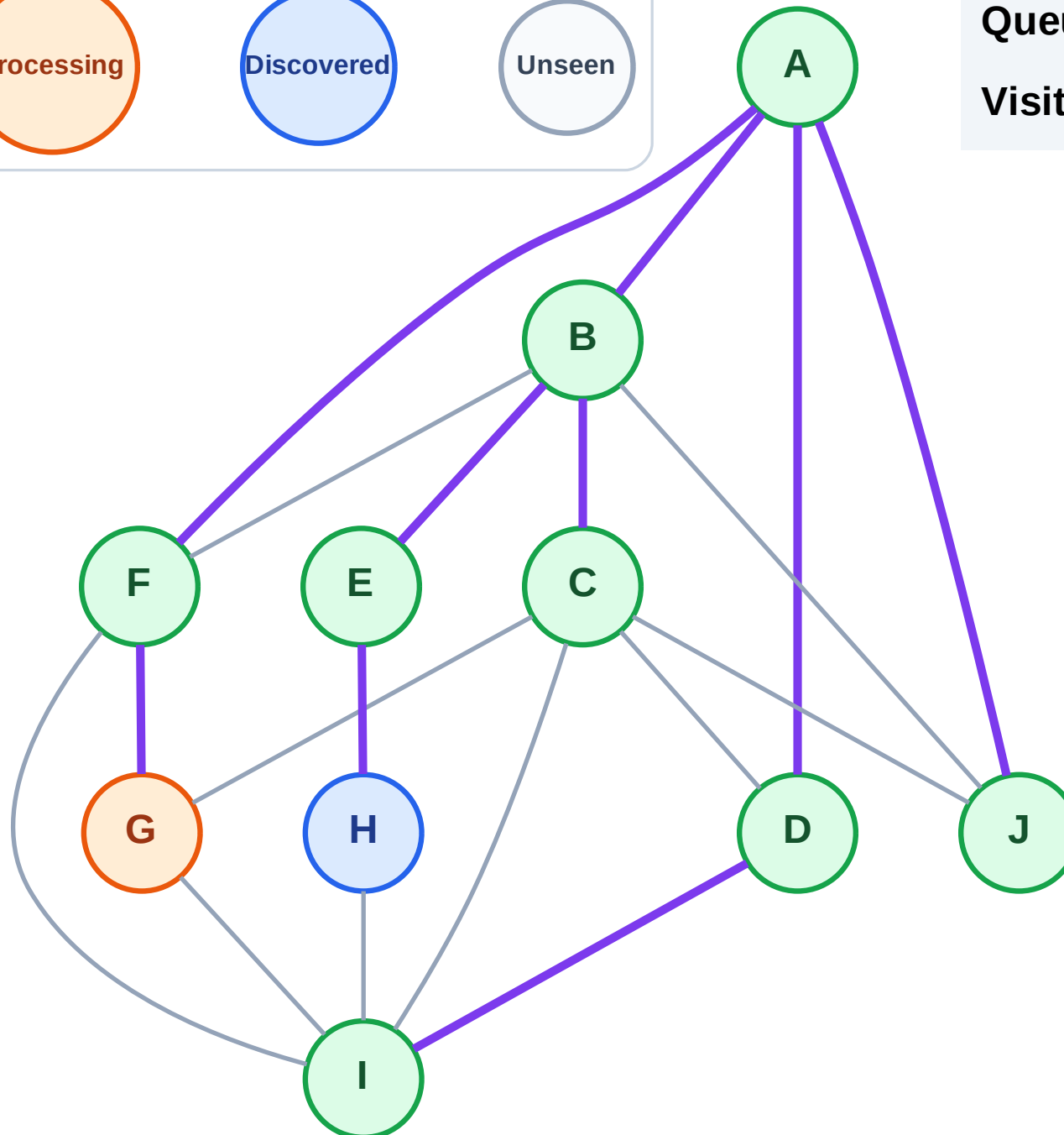
Step 18: Process node G

Legend



Queue: H

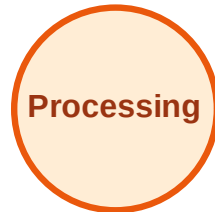
Visited: A, B, C, D, E, F, I, J



BFS Traversal

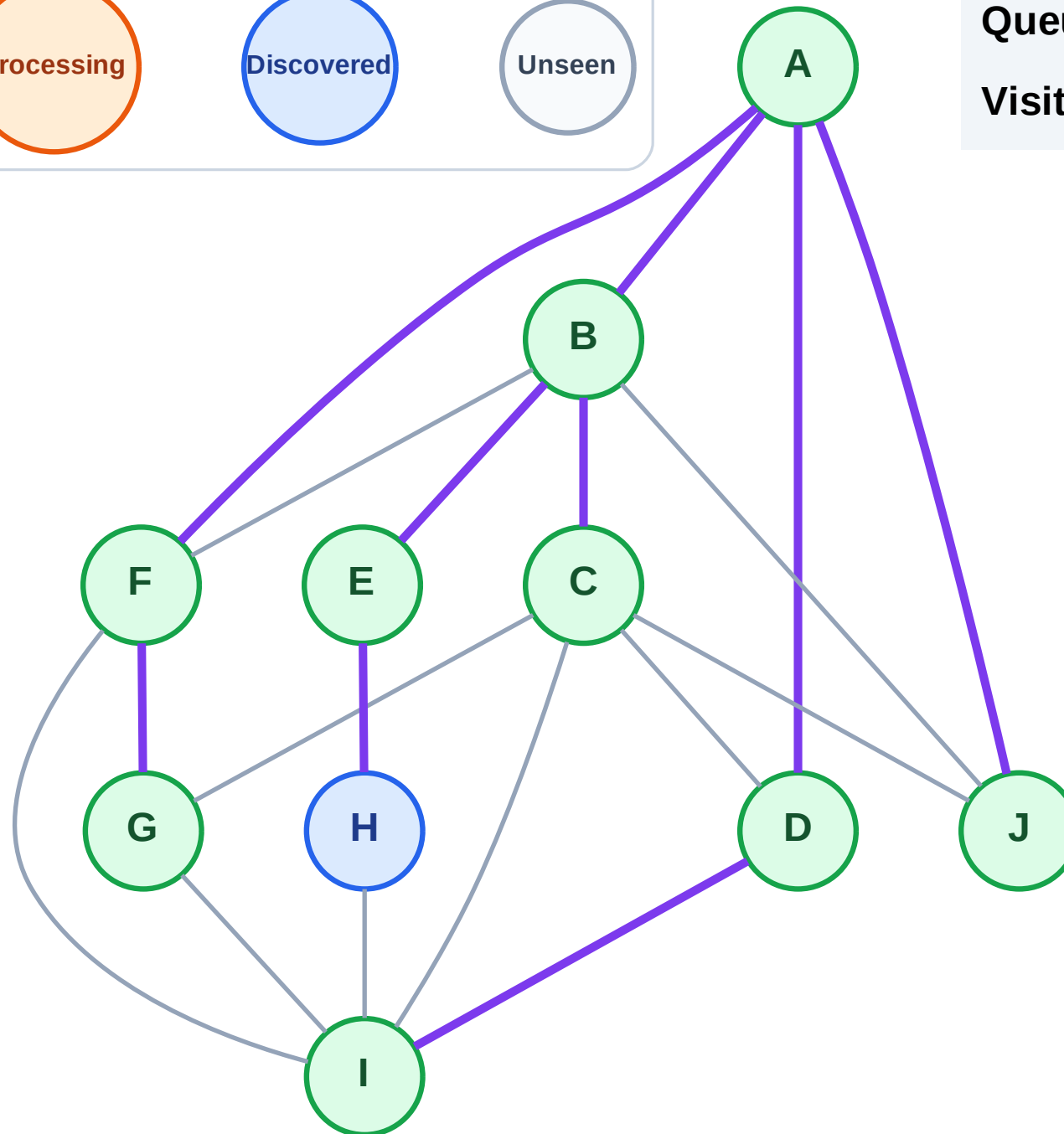
Step 19: No new neighbors from G

Legend



Queue: H

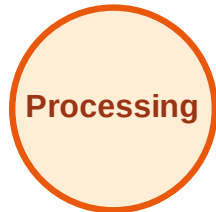
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

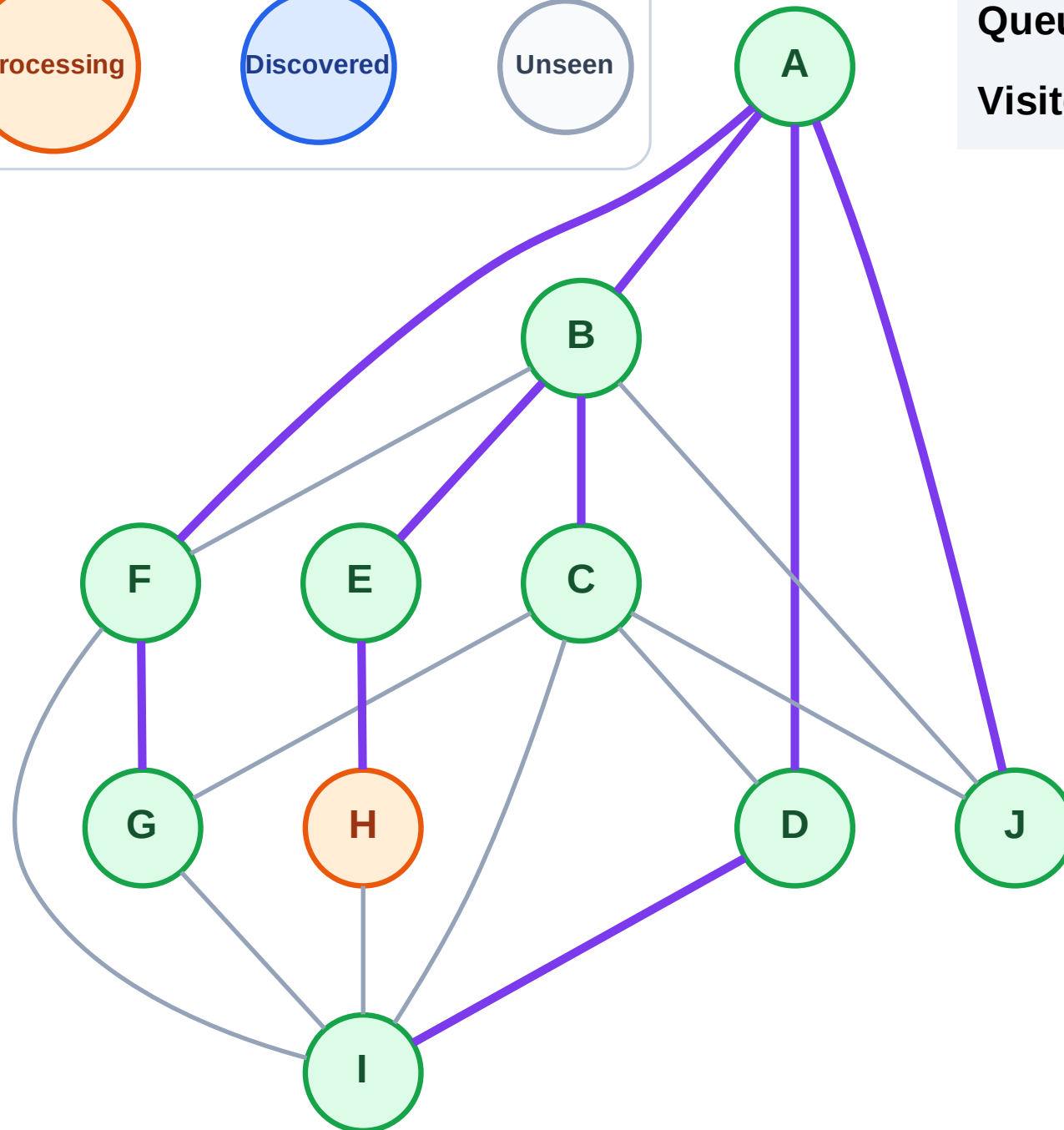
Step 20: Process node H

Legend



Queue: (empty)

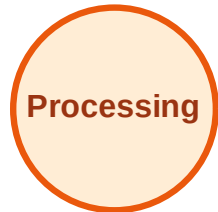
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

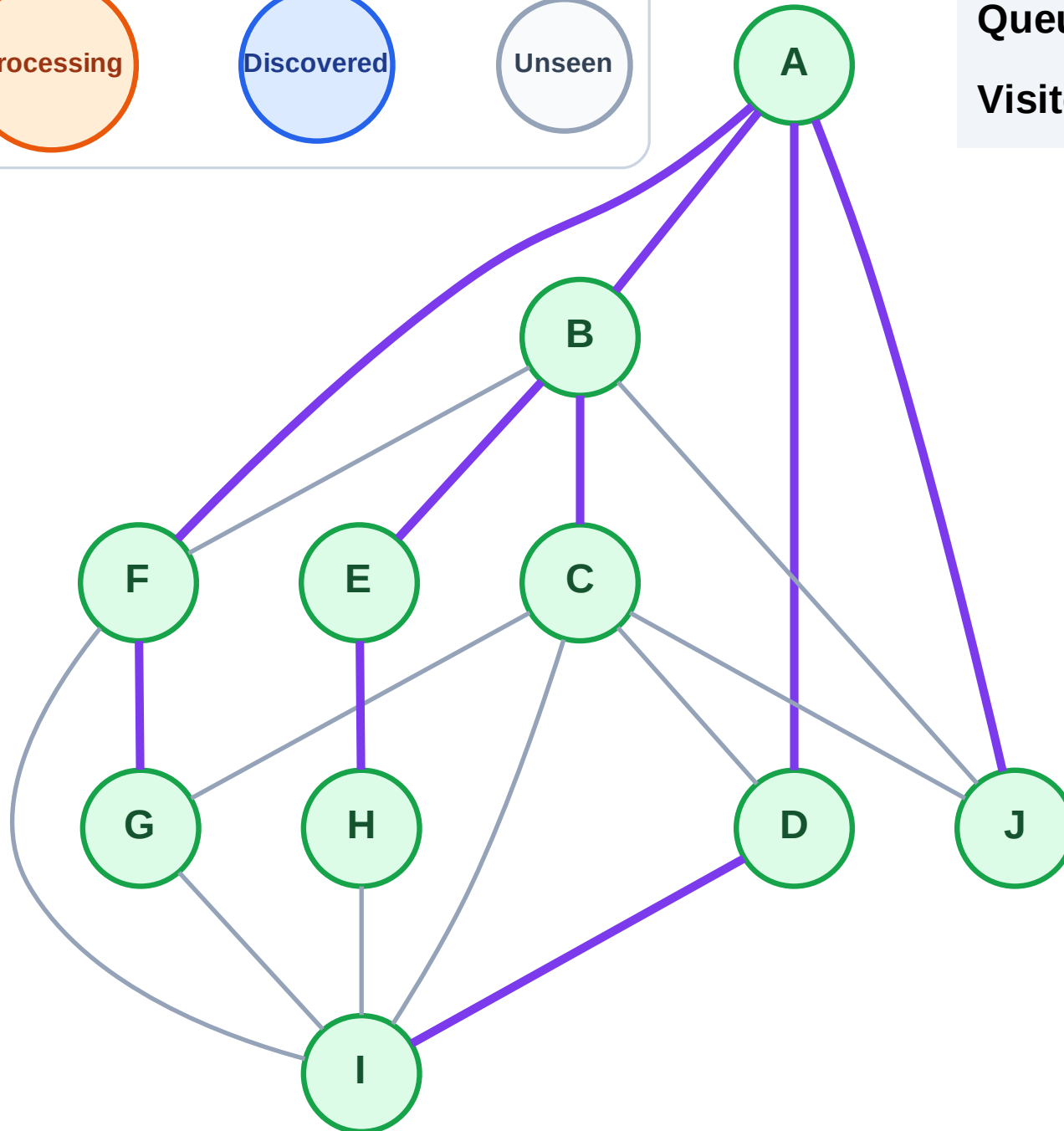
Step 21: No new neighbors from H

Legend



Queue: (empty)

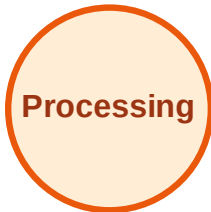
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

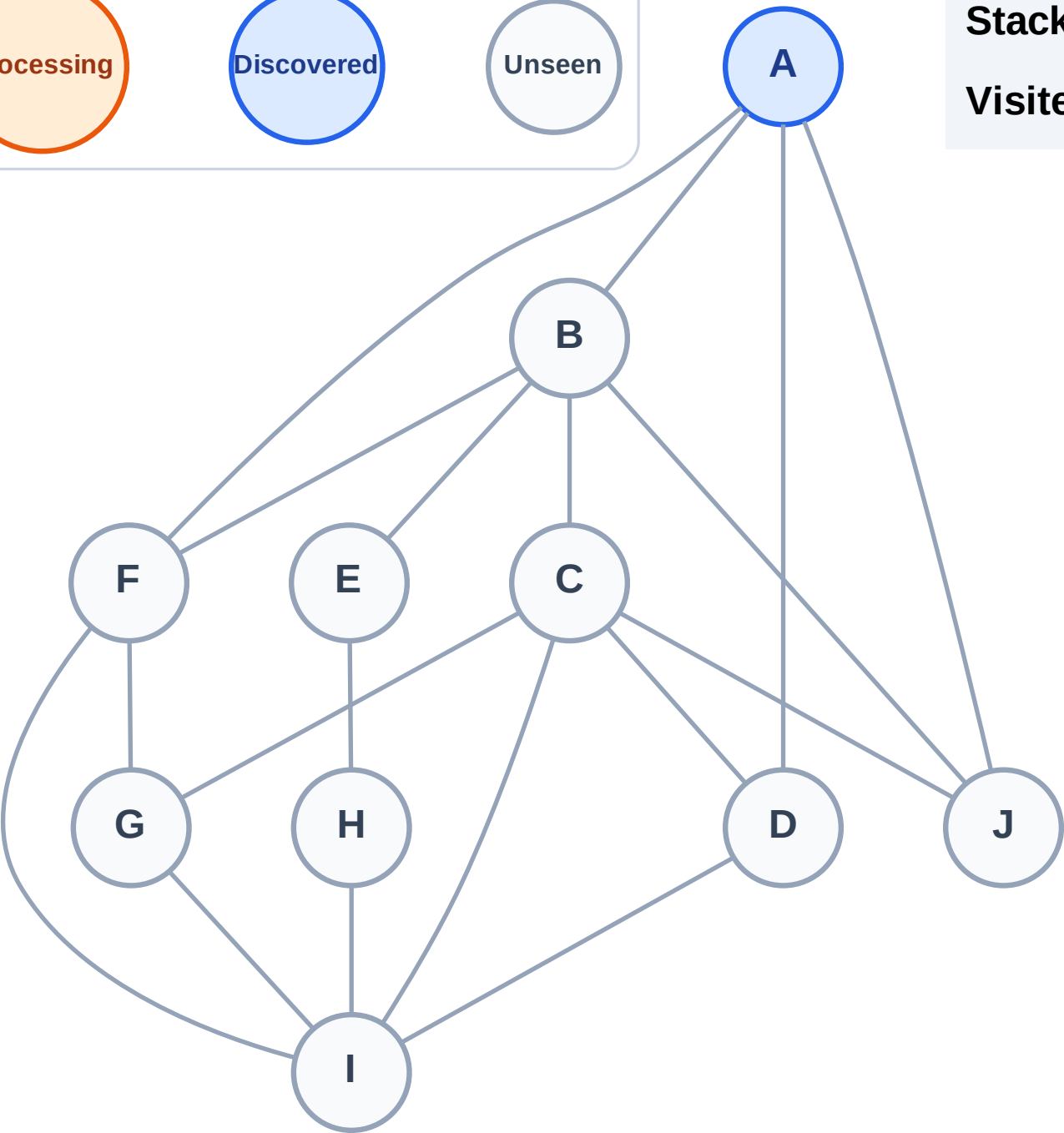
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

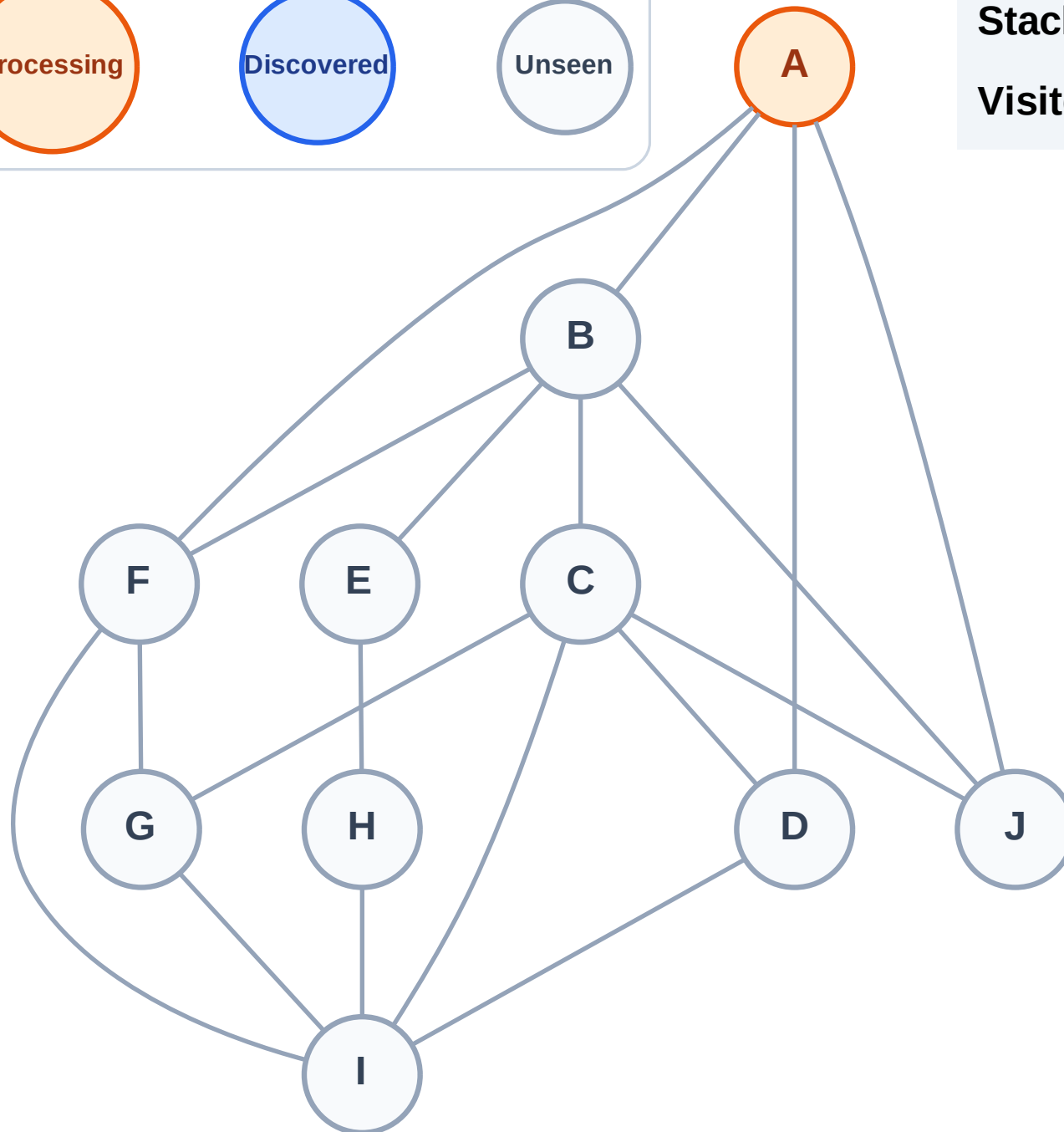
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

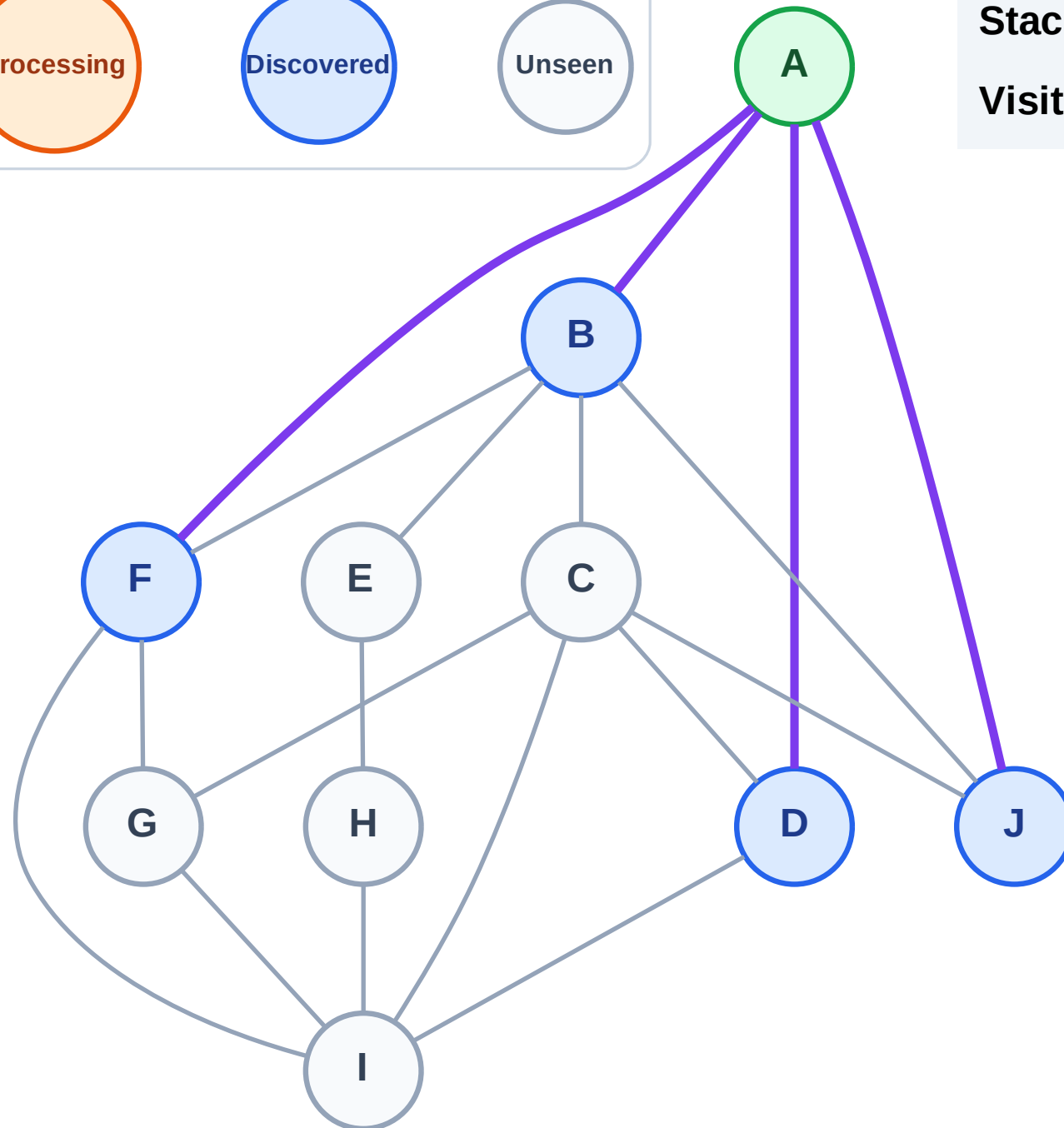
Step 3: Discover J, F, D, B from A

Legend



Stack (top at right): J | F | D | B

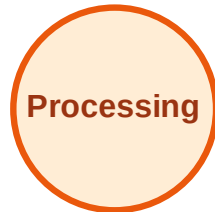
Visited: A



DFS Traversal

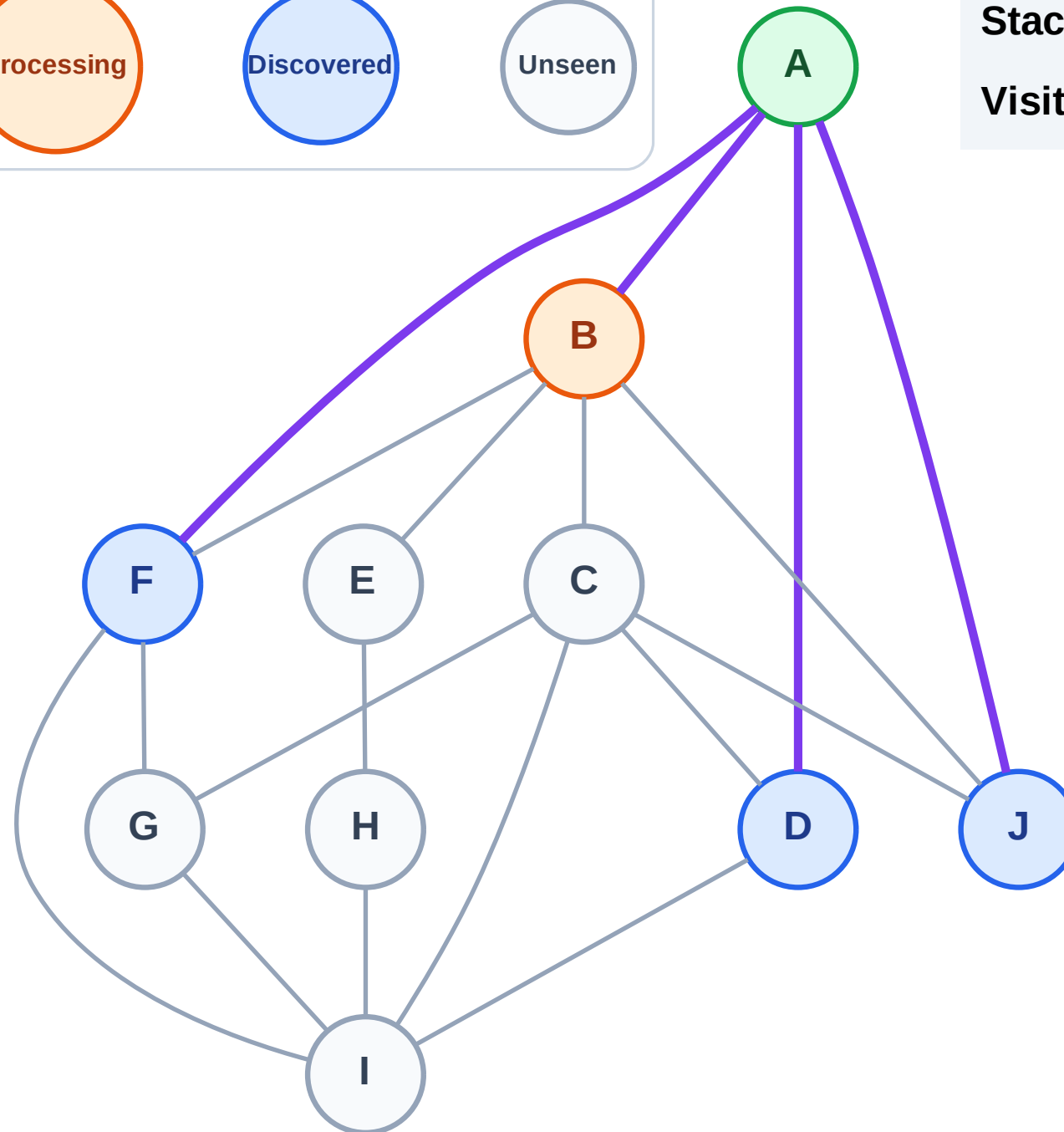
Step 4: Process node B

Legend



Stack (top at right): J | F | D

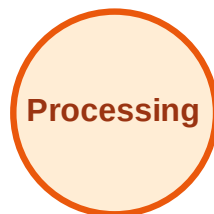
Visited: A



DFS Traversal

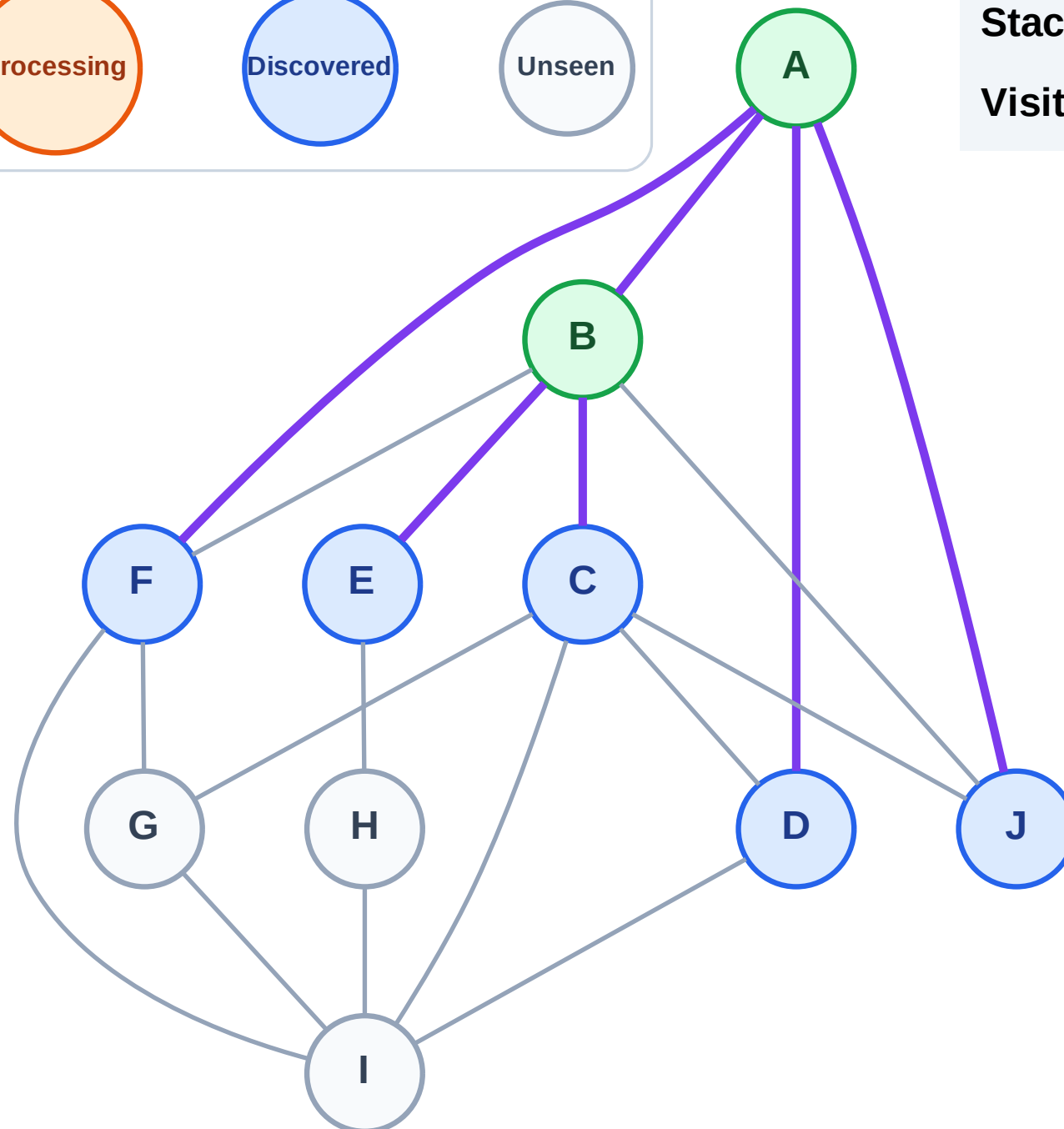
Step 5: Discover E, C from B

Legend



Stack (top at right): J | F | D | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

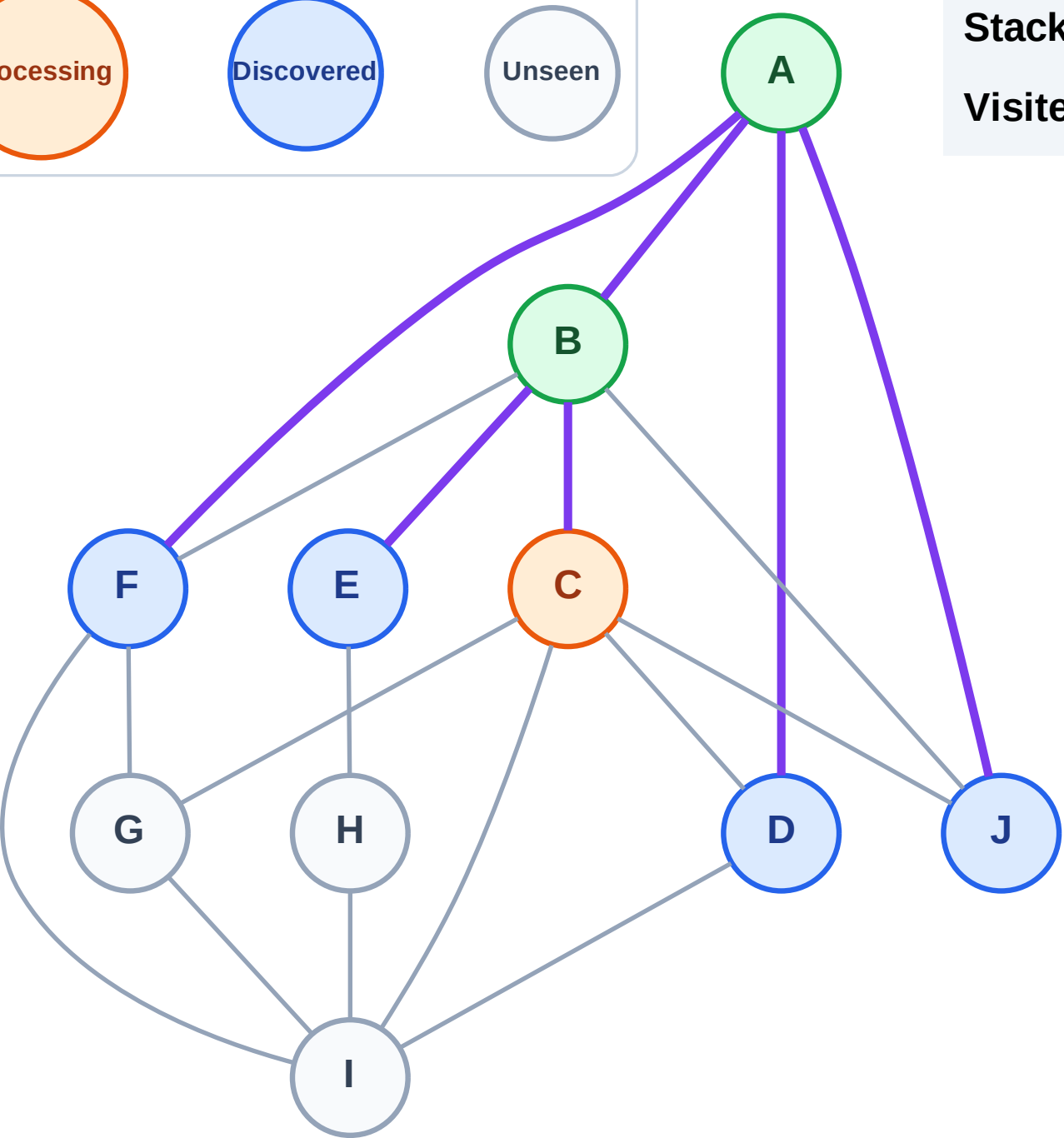
Processing

Discovered

Unseen

Stack (top at right): J | F | D | E

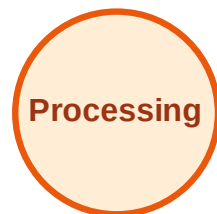
Visited: A, B



DFS Traversal

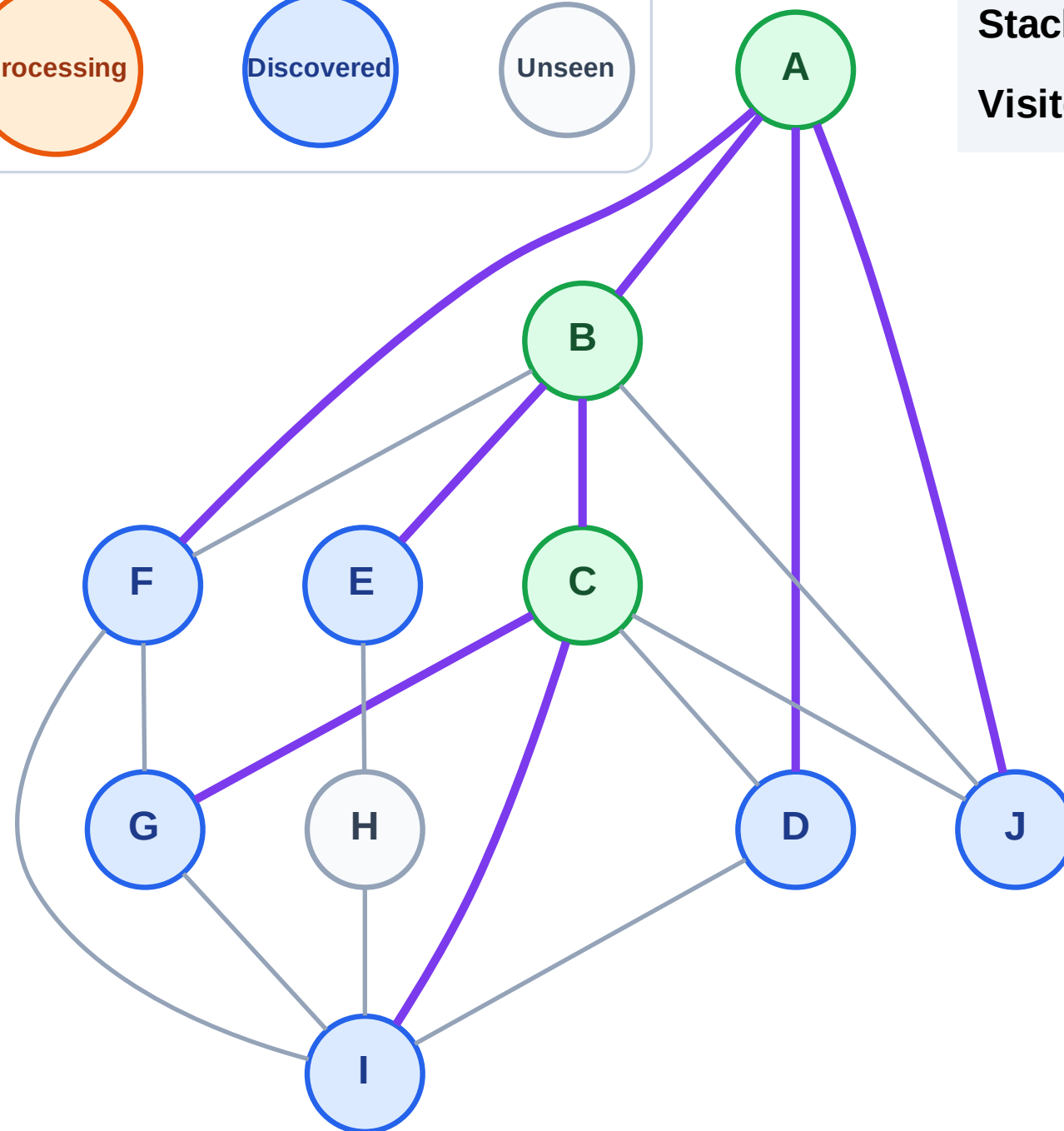
Step 7: Discover I, G from C

Legend



Stack (top at right): J | F | D | E | I | G

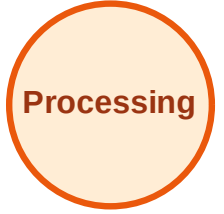
Visited: A, B, C



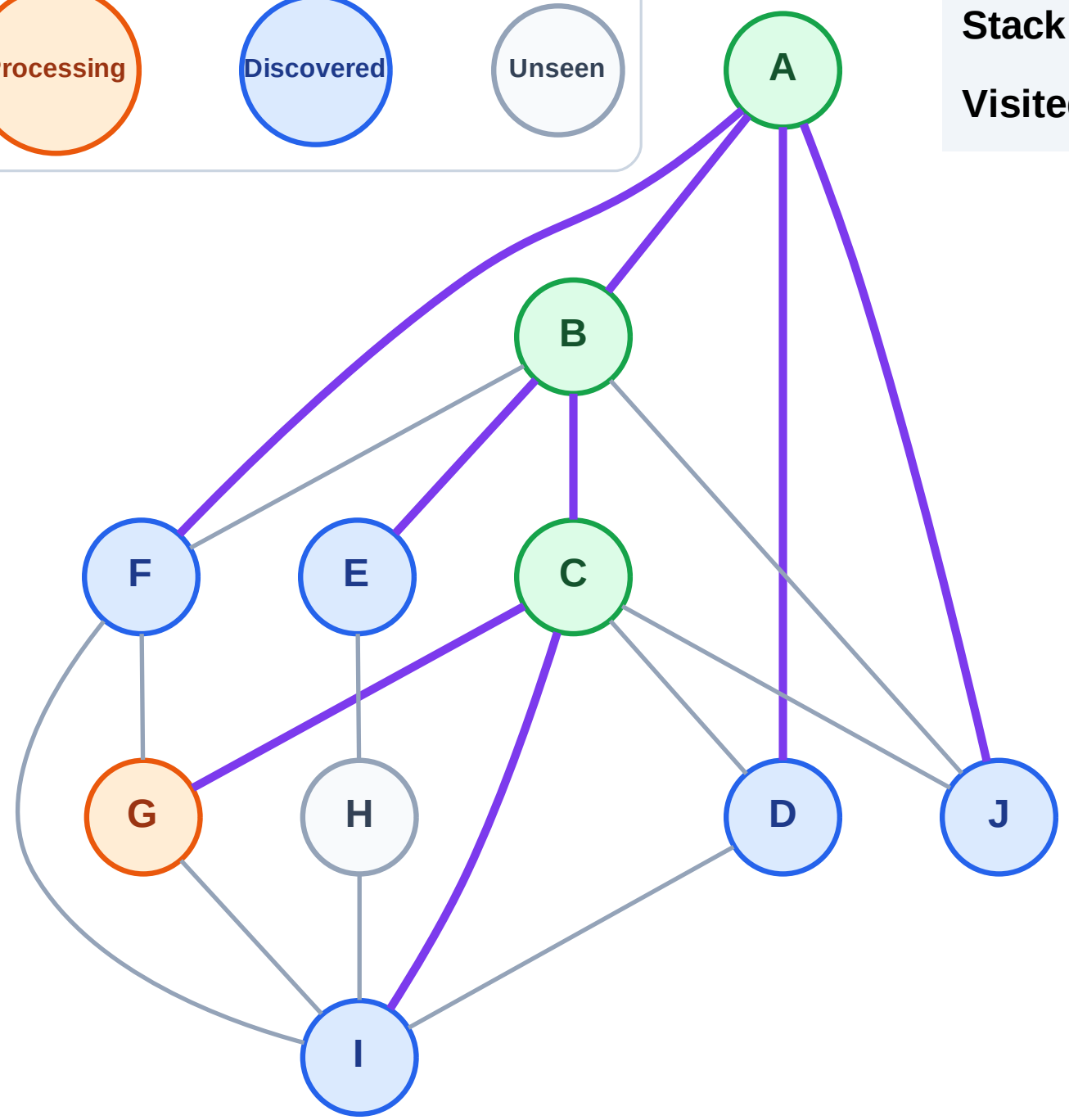
DFS Traversal

Step 8: Process node G

Legend



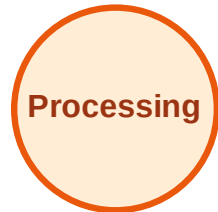
Stack (top at right): J | F | D | E | I
Visited: A, B, C



DFS Traversal

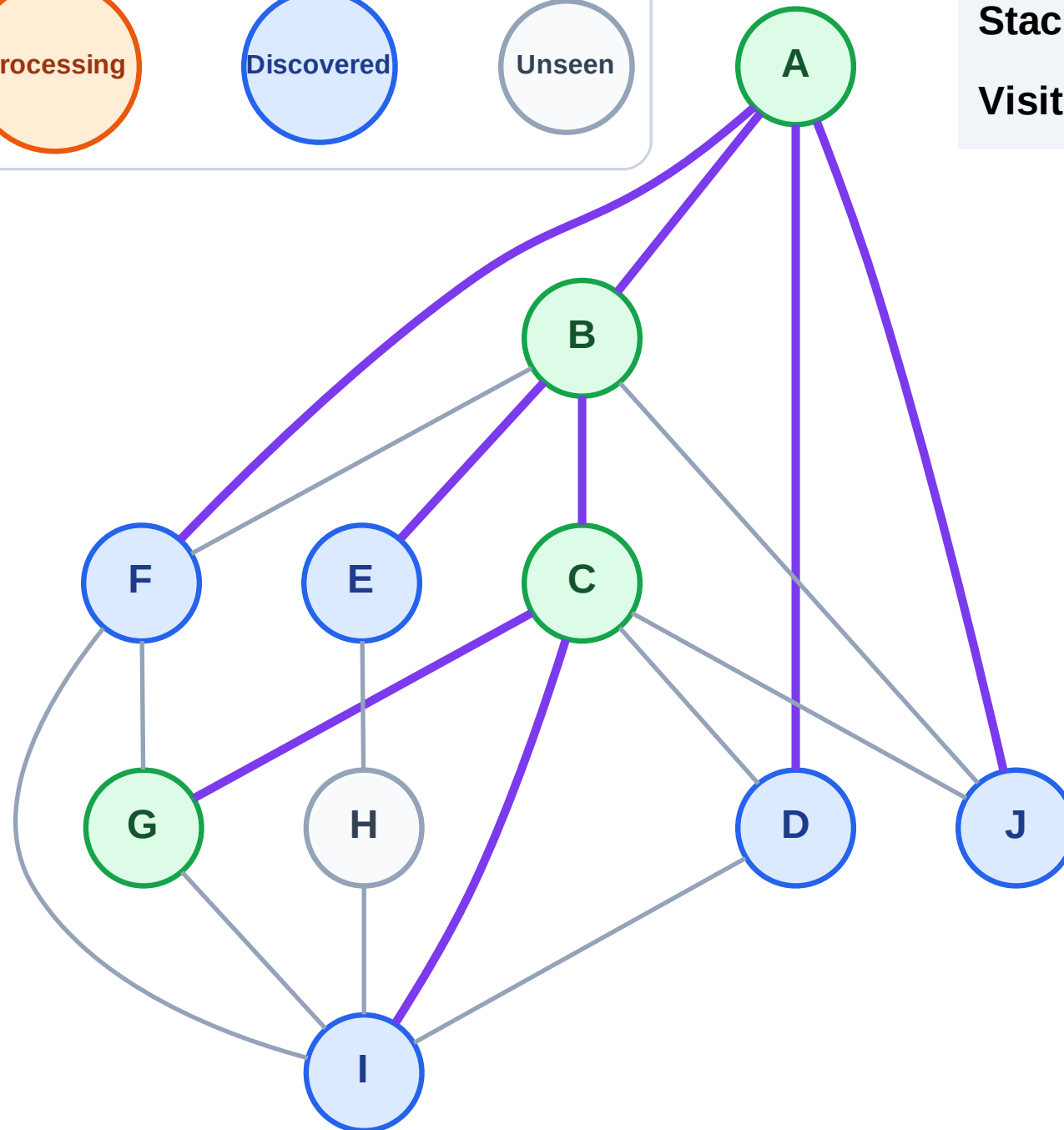
Step 9: No new neighbors from G

Legend



Stack (top at right): J | F | D | E | I

Visited: A, B, C, G



DFS Traversal

Step 10: Process node I

Legend

Complete

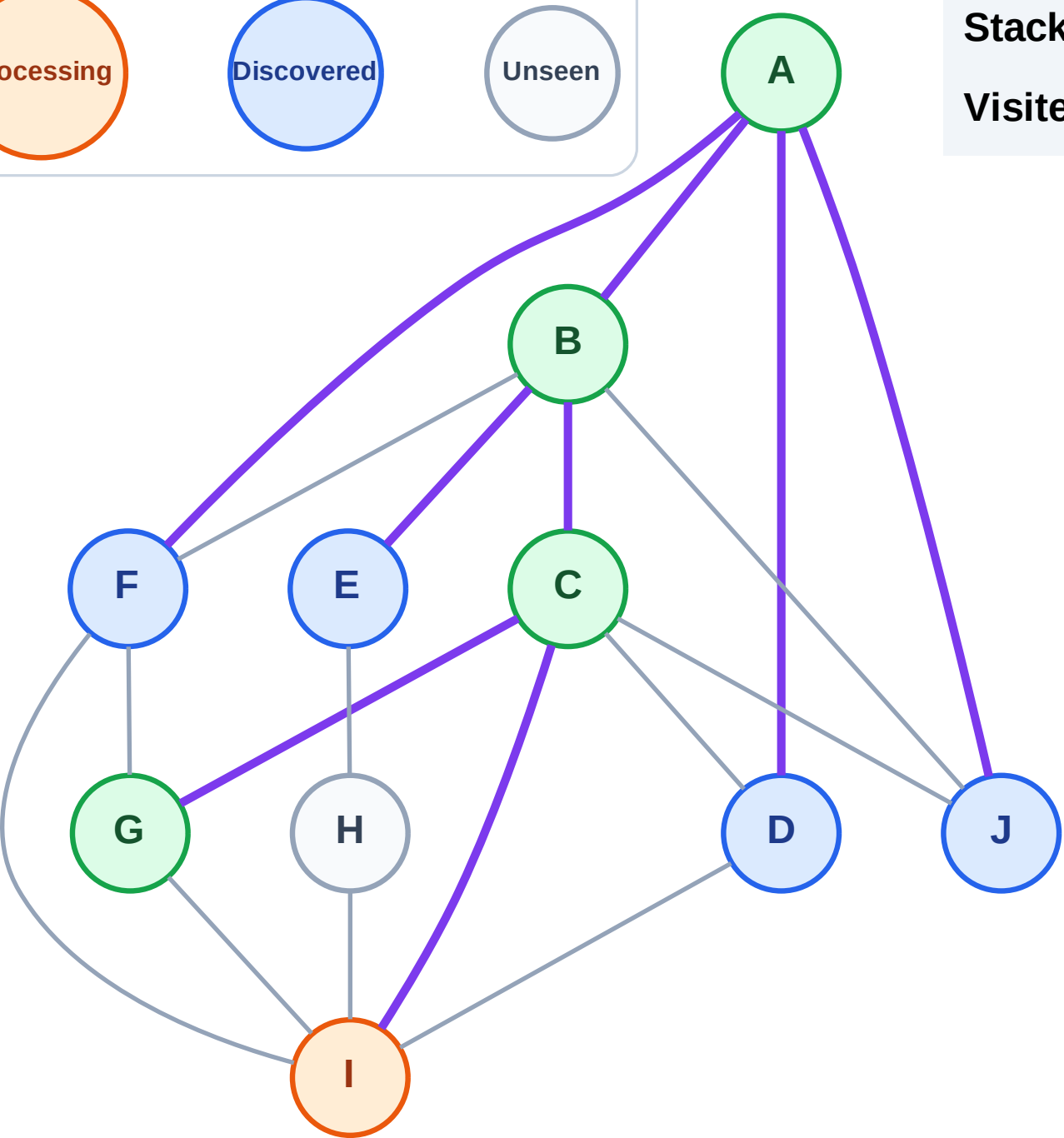
Processing

Discovered

Unseen

Stack (top at right): J | F | D | E

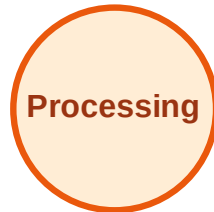
Visited: A, B, C, G



DFS Traversal

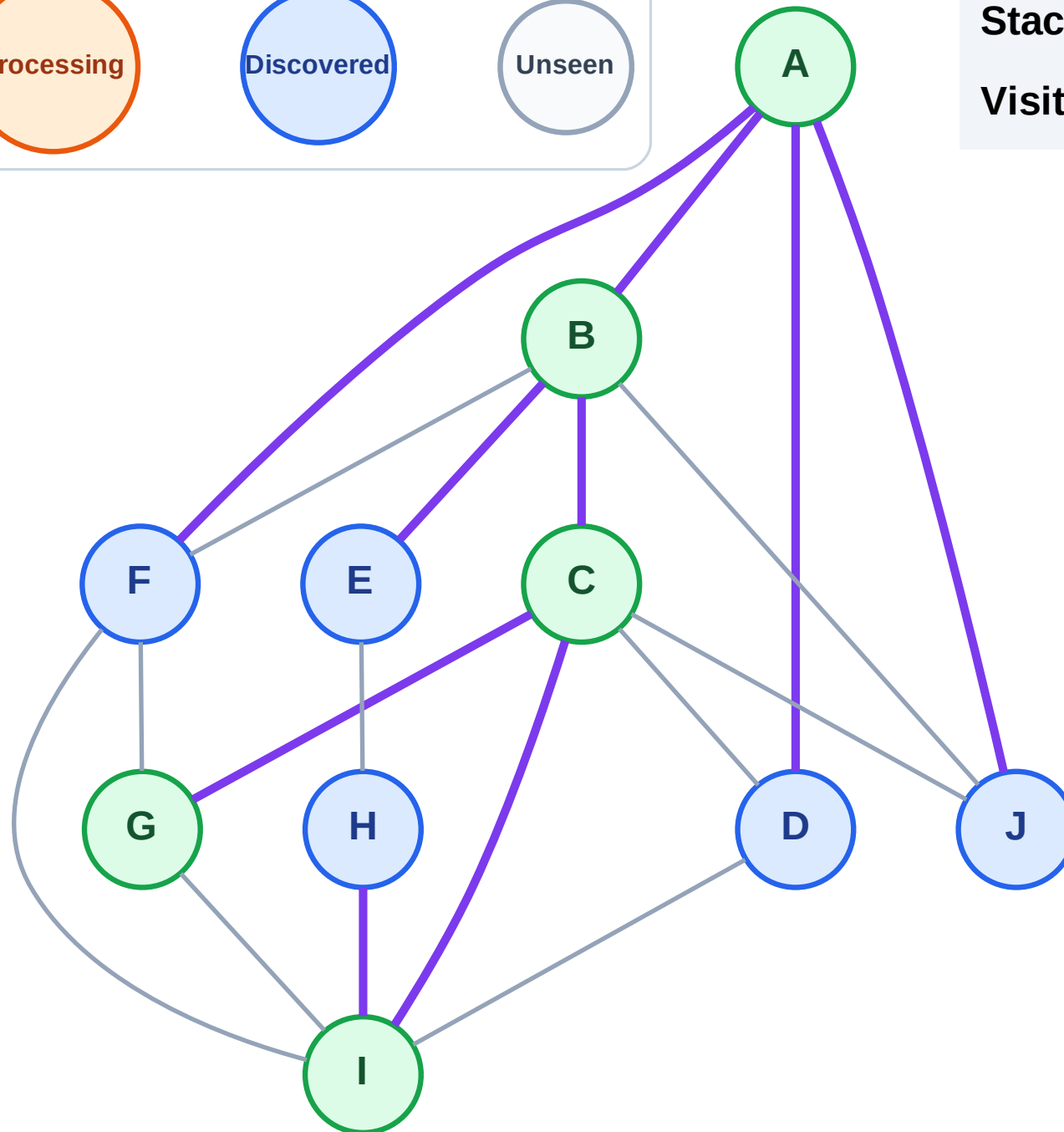
Step 11: Discover H from I

Legend



Stack (top at right): J | F | D | E | H

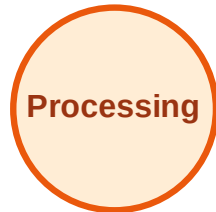
Visited: A, B, C, G, I



DFS Traversal

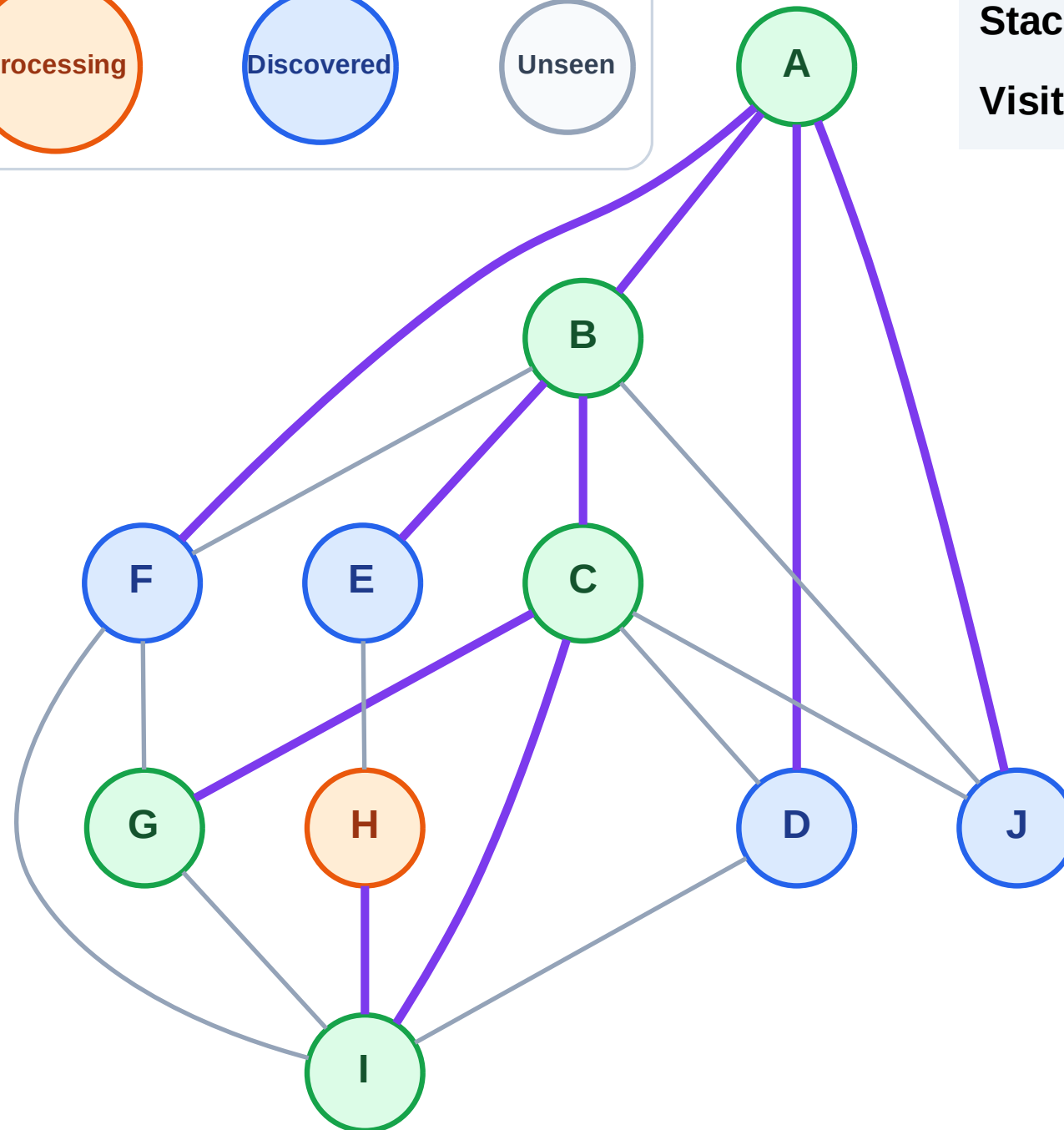
Step 12: Process node H

Legend



Stack (top at right): J | F | D | E

Visited: A, B, C, G, I



DFS Traversal

Step 13: No new neighbors from H

Legend

Complete

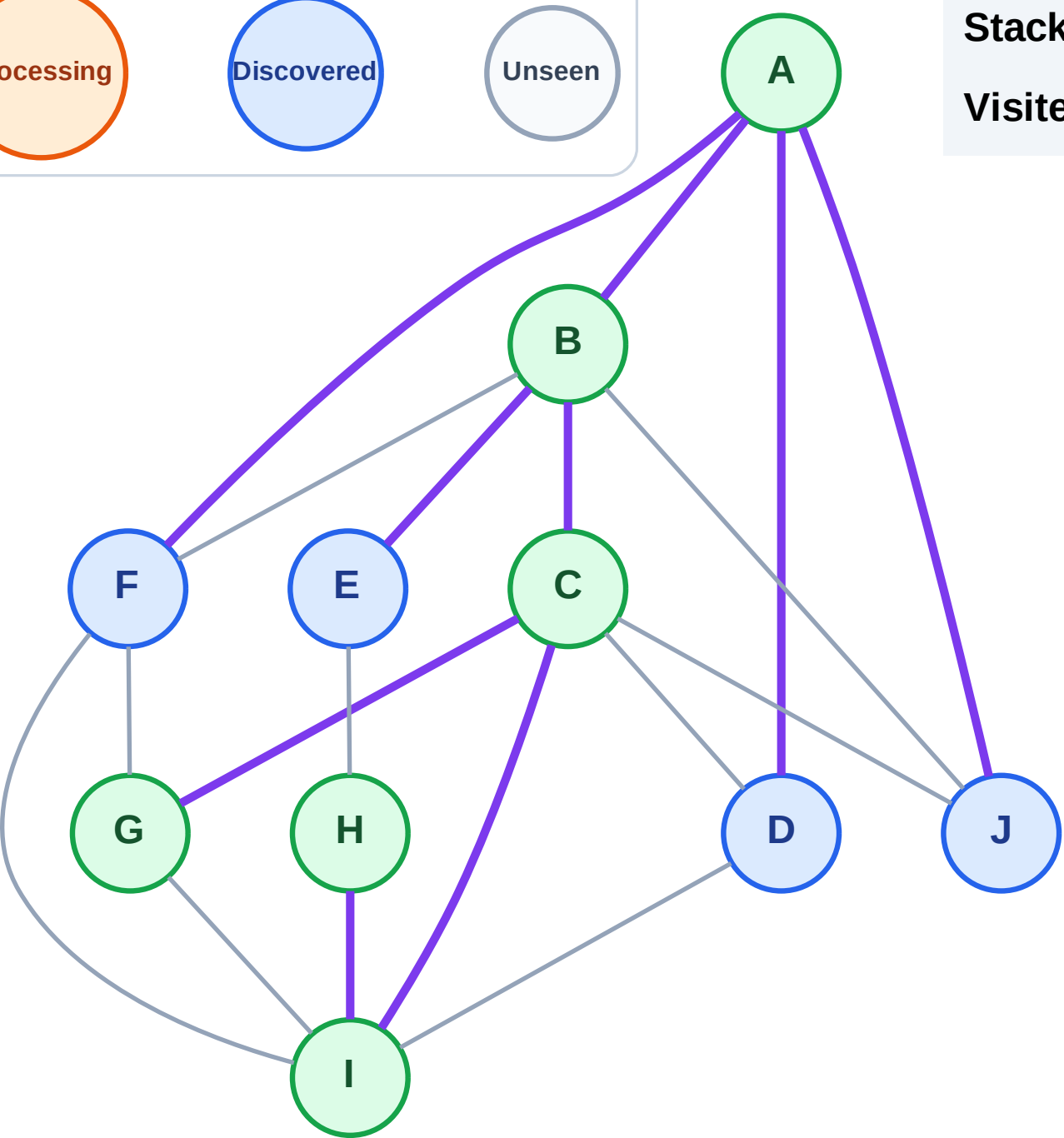
Processing

Discovered

Unseen

Stack (top at right): J | F | D | E

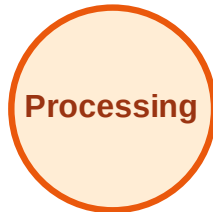
Visited: A, B, C, G, H, I



DFS Traversal

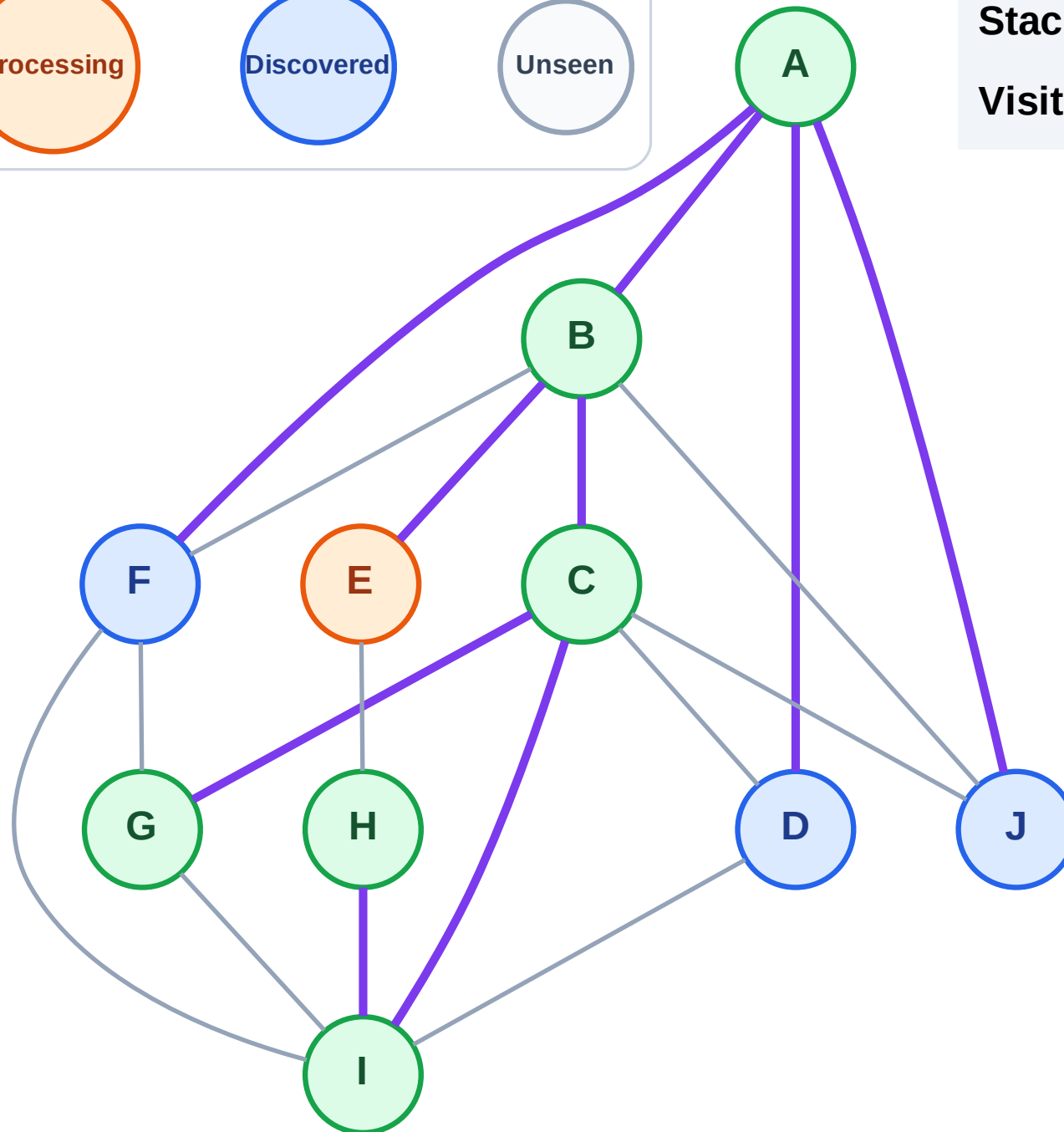
Step 14: Process node E

Legend



Stack (top at right): J | F | D

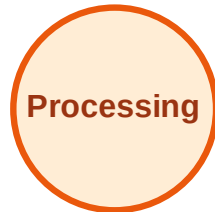
Visited: A, B, C, G, H, I



DFS Traversal

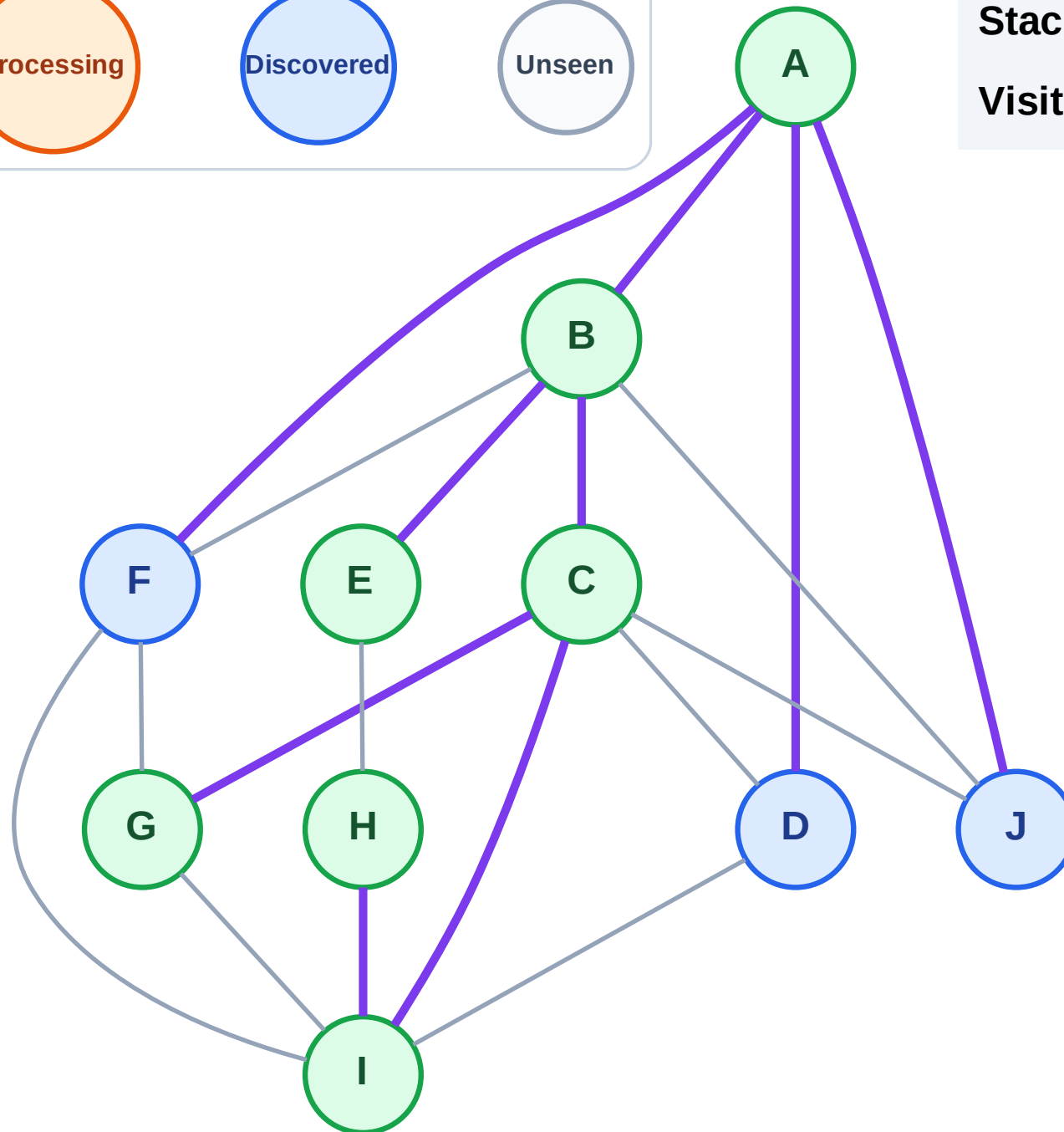
Step 15: No new neighbors from E

Legend



Stack (top at right): J | F | D

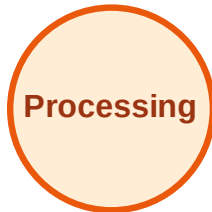
Visited: A, B, C, E, G, H, I



DFS Traversal

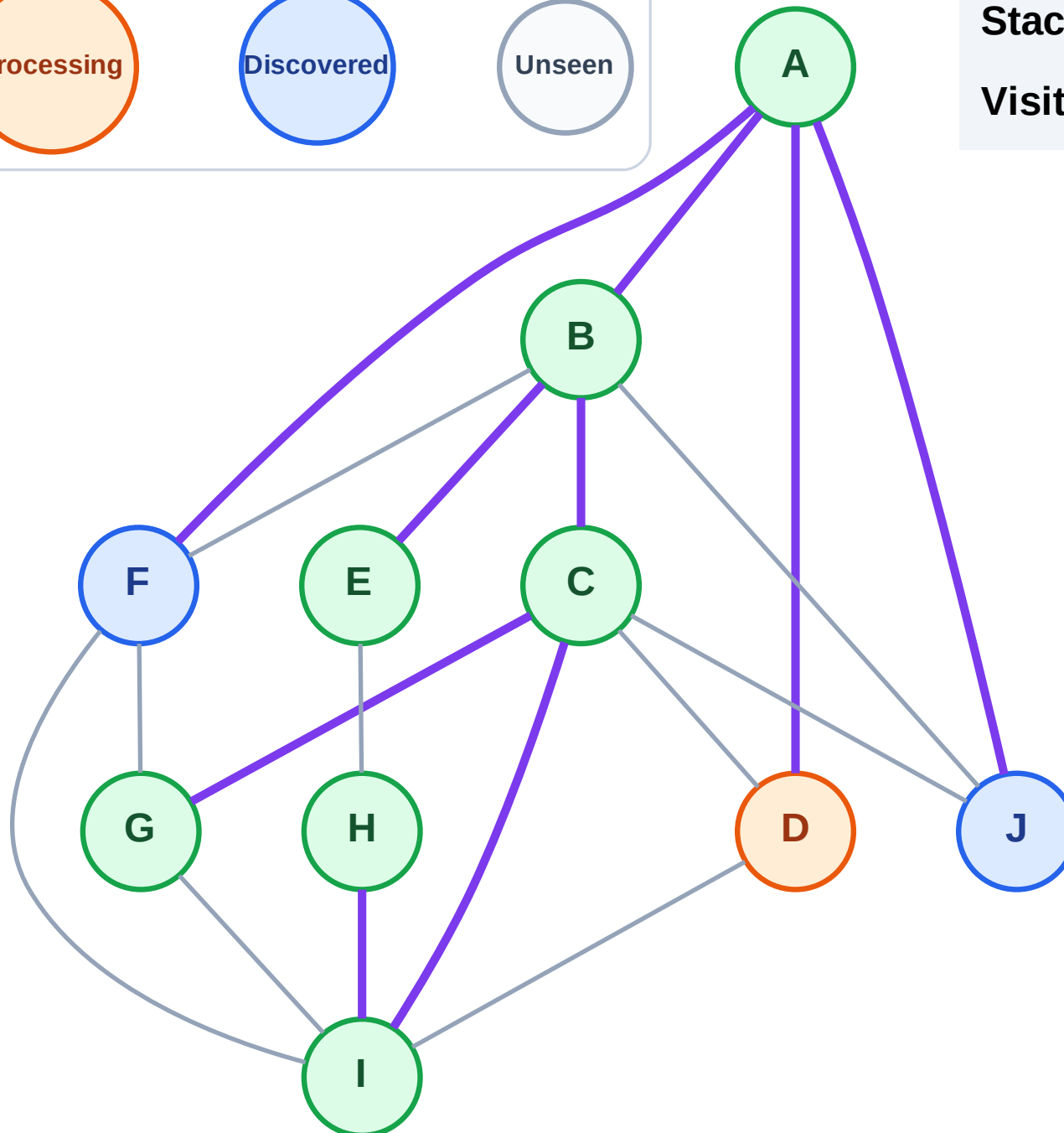
Step 16: Process node D

Legend



Stack (top at right): J | F

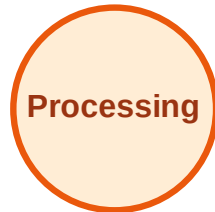
Visited: A, B, C, E, G, H, I



DFS Traversal

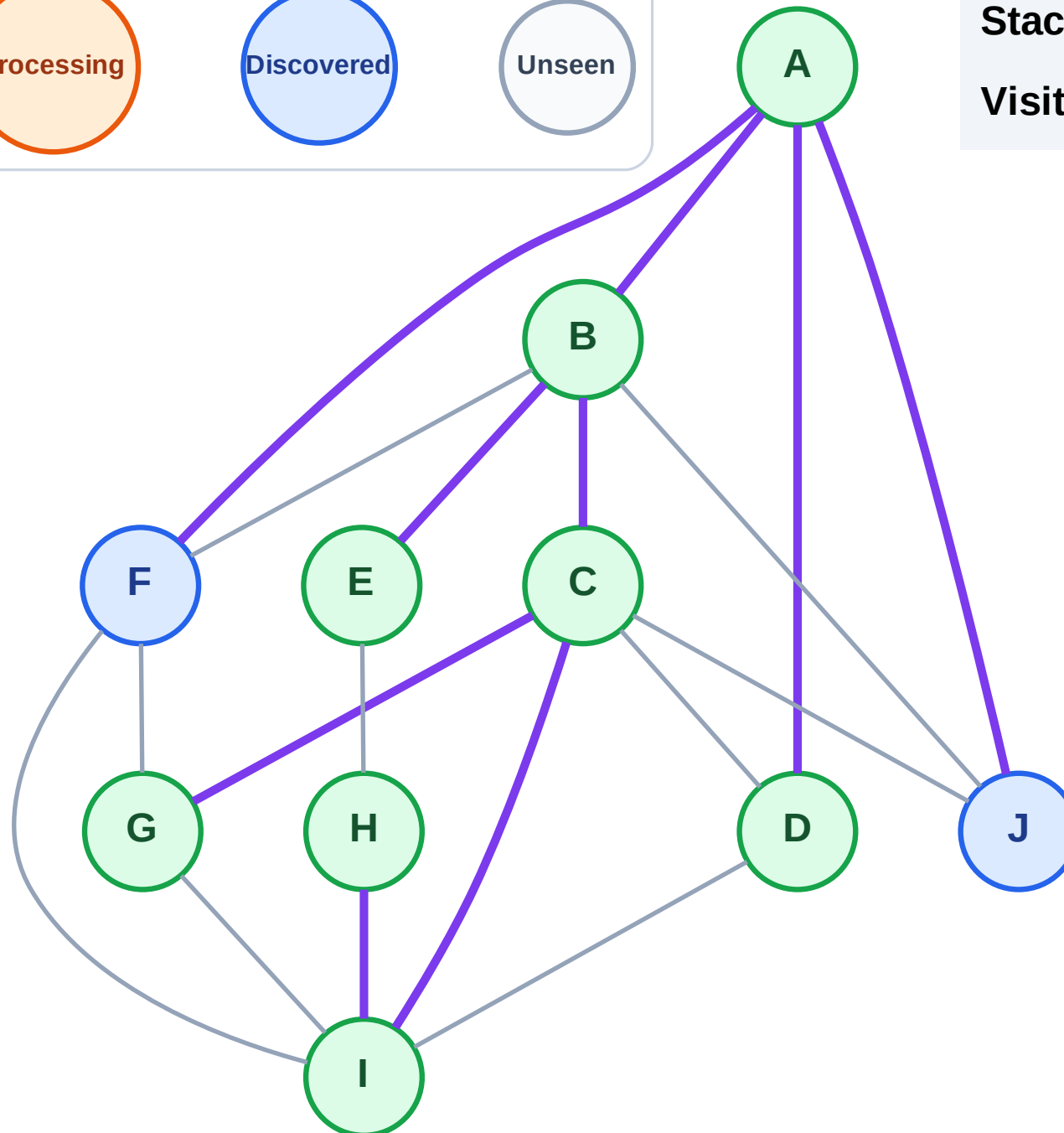
Step 17: No new neighbors from D

Legend



Stack (top at right): J | F

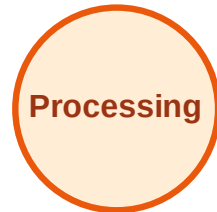
Visited: A, B, C, D, E, G, H, I



DFS Traversal

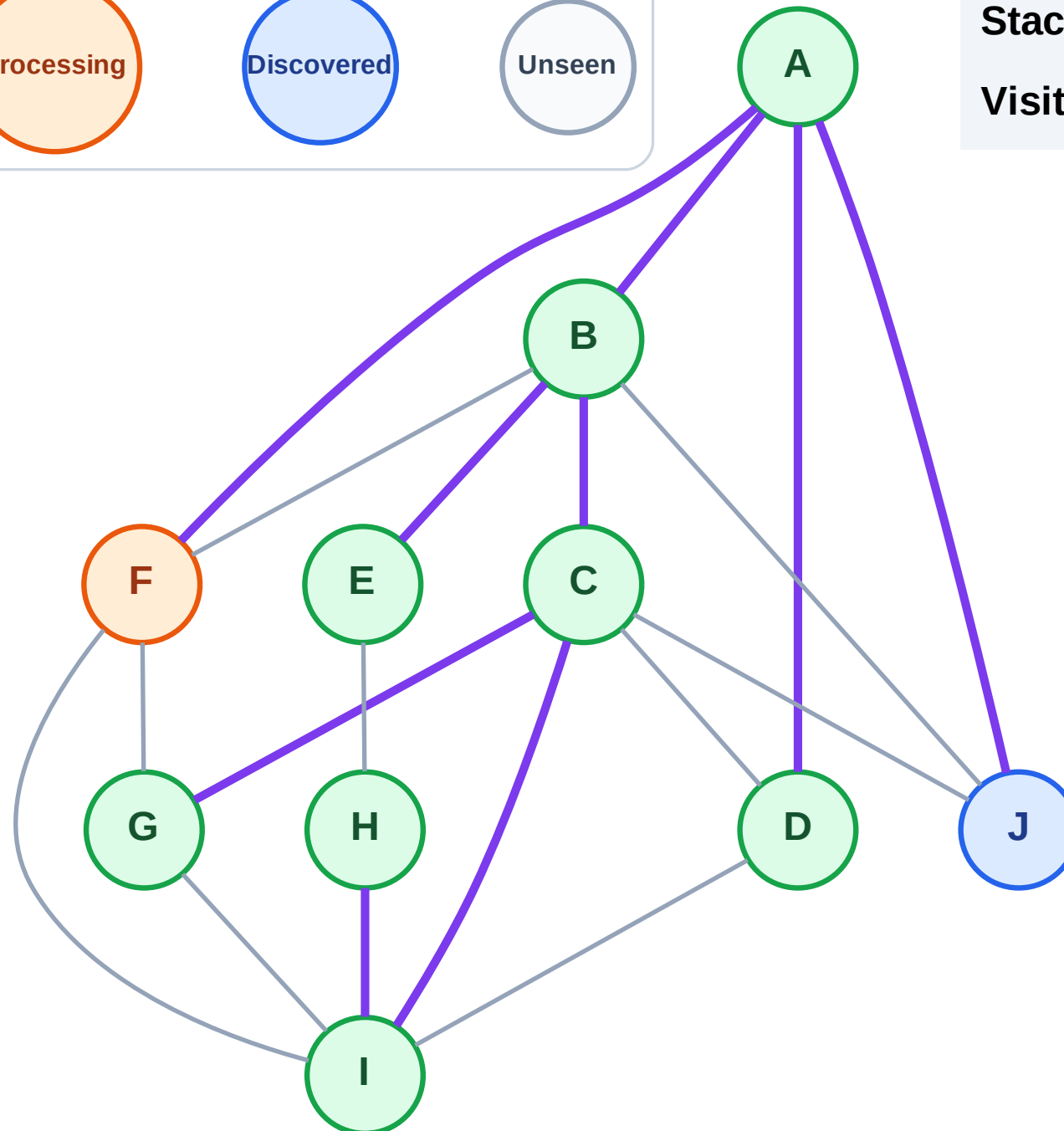
Step 18: Process node F

Legend



Stack (top at right): J

Visited: A, B, C, D, E, G, H, I

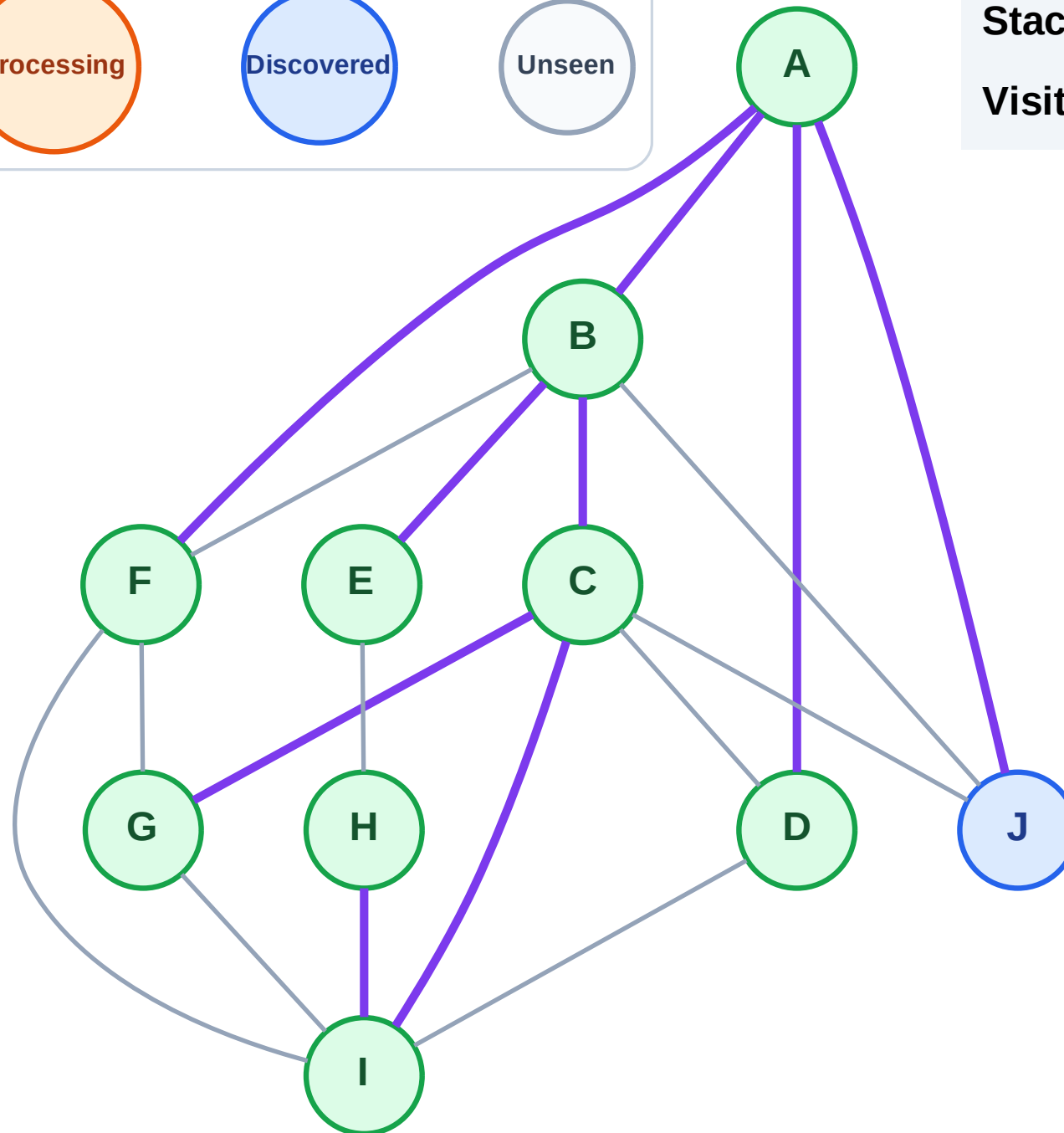


Step 19: No new neighbors from F

Legend



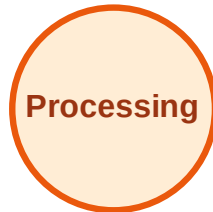
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

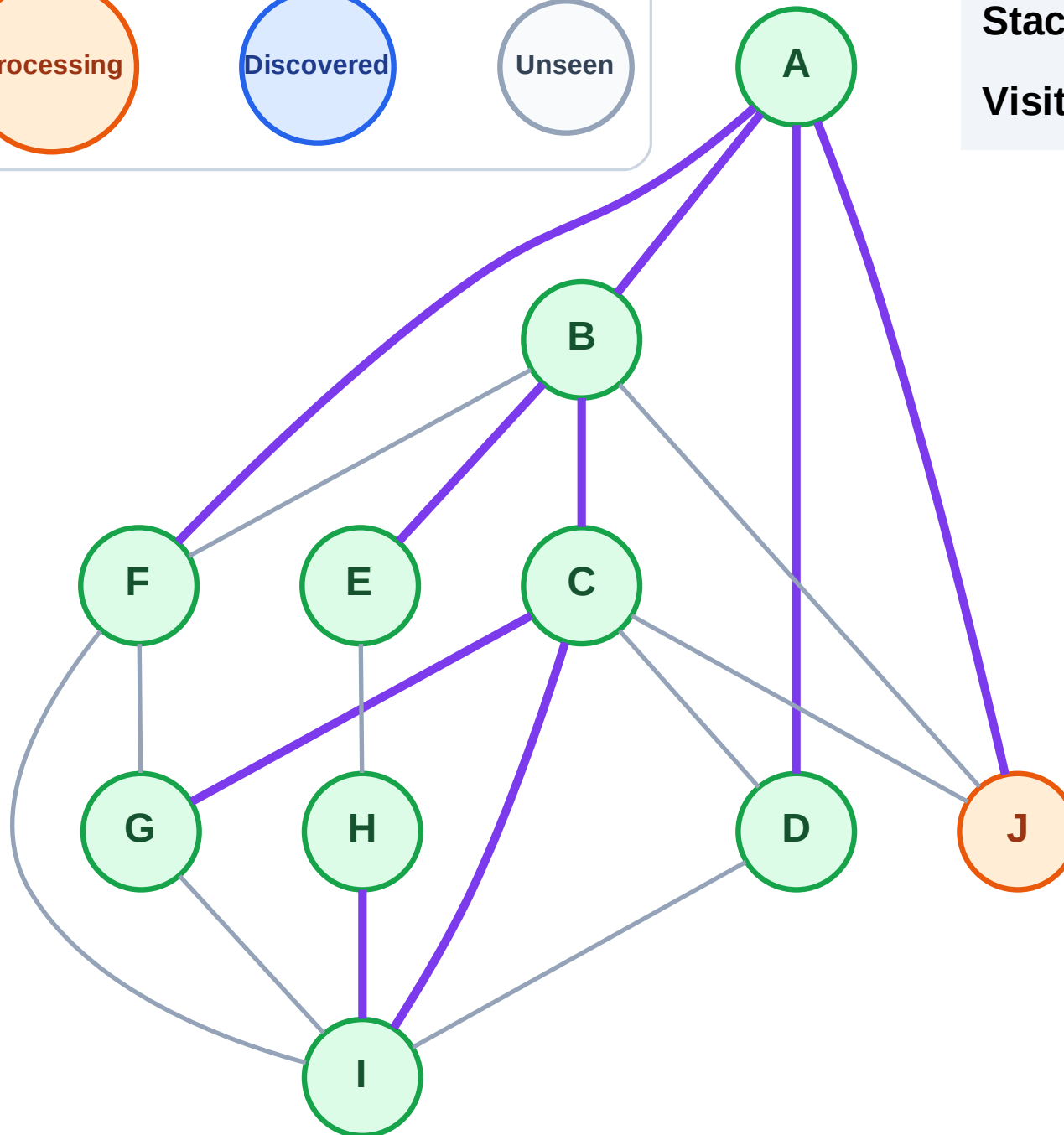
Step 20: Process node J

Legend



Stack (top at right): (empty)

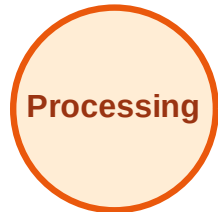
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

